

McStas Union components

What they are and how to use them

PRESENTED BY MADS BERTELSEN

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This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 823852

Introduction

Who am I?

- Worked with McStas last 9 years, now McStas developer
- PhD from University of Copenhagen under Kim Lefmann in 2017
- Now Post Doc at ESS Data Management and Software Centre, funded by PaNOSC and HighNESS
- Writes software, happy when someone wants to use it and learn about my projects!

Mads Bertelsen



guide_bot Automatic guide optimization given requirements and overall geometry
(MATLAB)

Union components Modular samples, multiple scattering within
McStas

McStasScript Flexible McStas Python API



Introduction

Difficulties with realistic samples

- Sample holders with complicated geometry
- Consists of many different materials
- No clear order in which neutrons would pass through
- Could have co aligned crystals
- Twinned crystals
- It could all be in a sample environment



Photo by Pia Ray Jensen

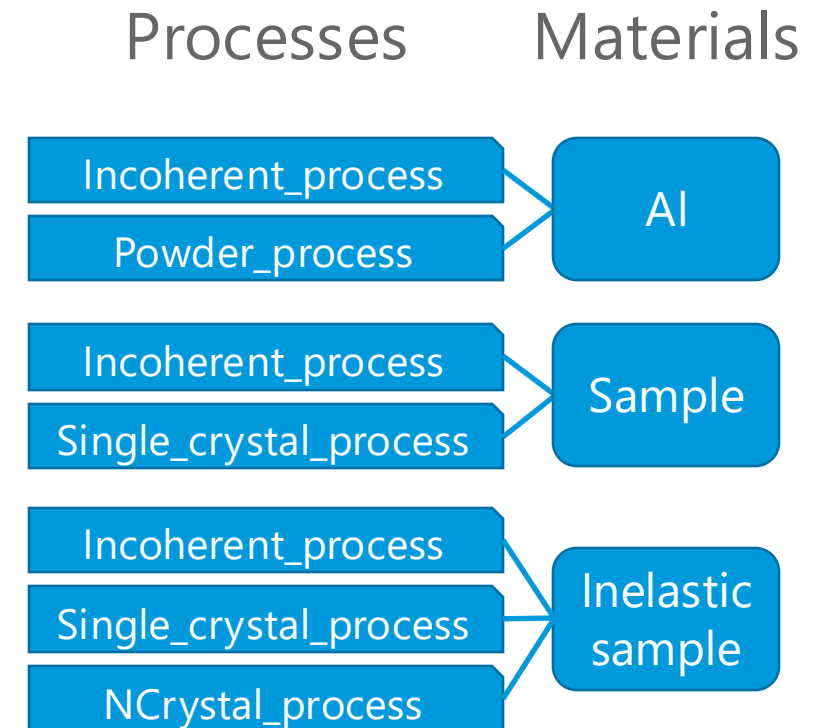


Physics

Modular physics description



- Combine simple physical processes
 - Incoherent scattering
 - Powder Bragg scattering
 - Single crystal Bragg scattering
 - Simple phonon
 - ...
- Create material definitions by combining an arbitrary number of these processes and an absorption cross section



Physics

Modular physics description

```
COMPONENT Al_incoherent = Incoherent_process(  
  sigma = 4*0.0082, packing_factor=1,  
  unit_cell_volume = 66.4)  
AT (0,0,0) ABSOLUTE
```

```
COMPONENT Al_powder = Powder_process(  
  reflections = "Al.laz")  
AT (0,0,0) ABSOLUTE
```

```
COMPONENT Al = Union_make_material(  
  my_absorption = 100*4*0.231/66.4,  
  process_string = "Al_incoherent,Al_powder")  
AT (0,0,0) ABSOLUTE
```



Processes

Materials



Absorption given as inverse penetration depth

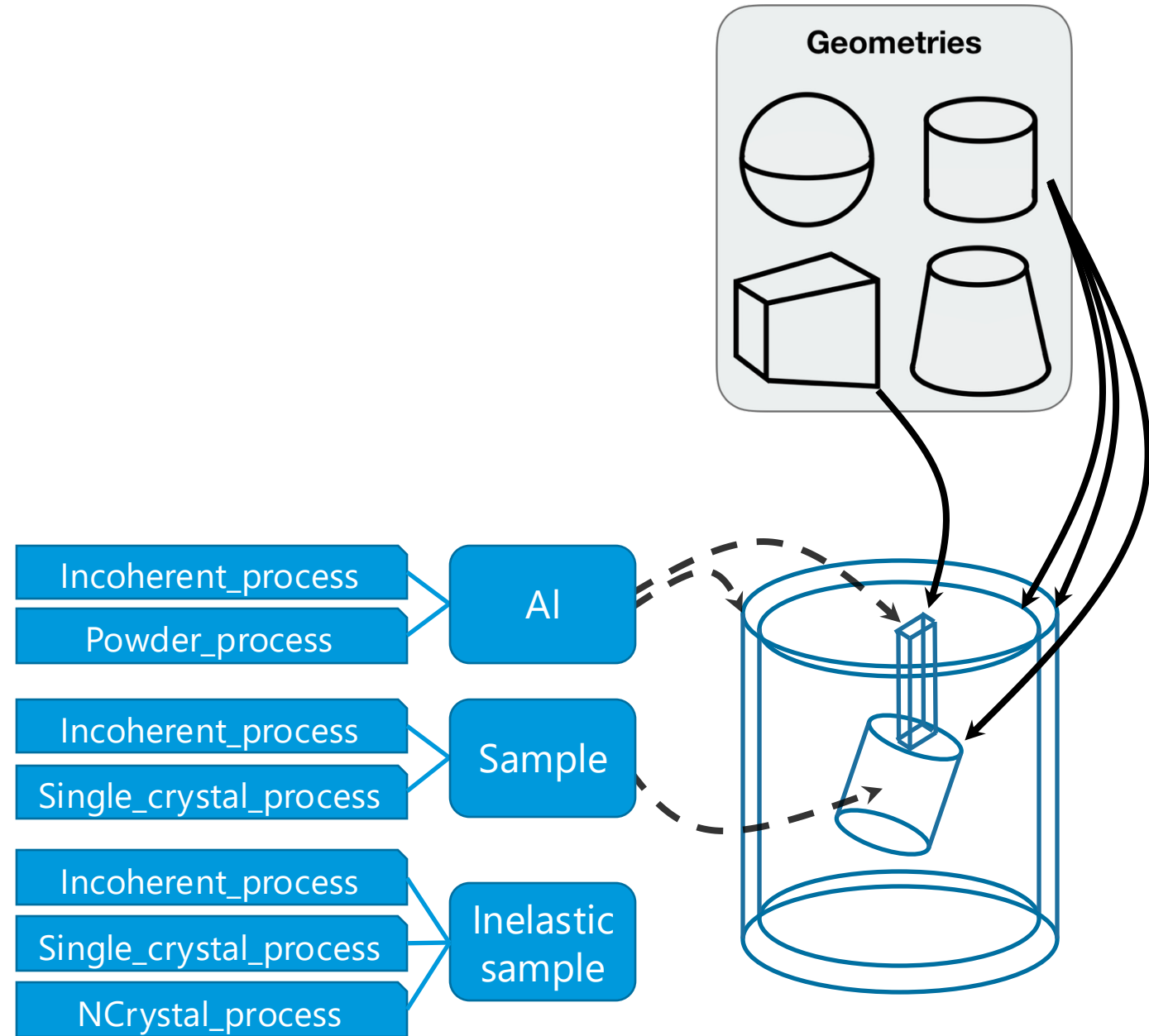
$$\text{my [1/m]} = \frac{\text{cross section per unit cell}}{\text{unit cell volume}}$$



Geometry

Modular geoemtry

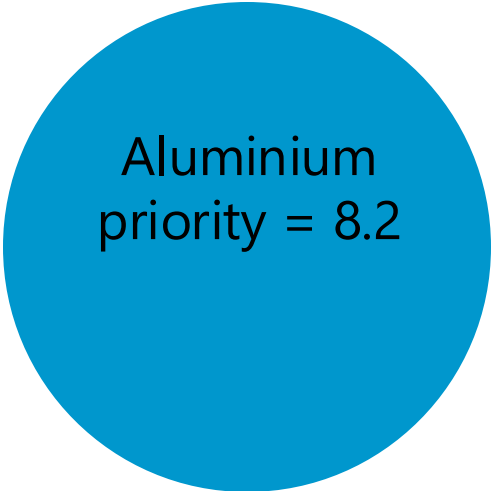
- Construct the desired geometry from simple basic shapes
 - Sphere
 - Cylinder
 - Box
 - Cone
 - CAD model (stl files)
- Assign each geometry a material



Geometry

Priority

- Each geometry is given a priority
- When two geometries overlap, the overlapping region has the material of the highest priority
- This allows the user to create complex geometries

A large blue circle representing a geometry with a priority of 8.2.

Aluminium
priority = 8.2

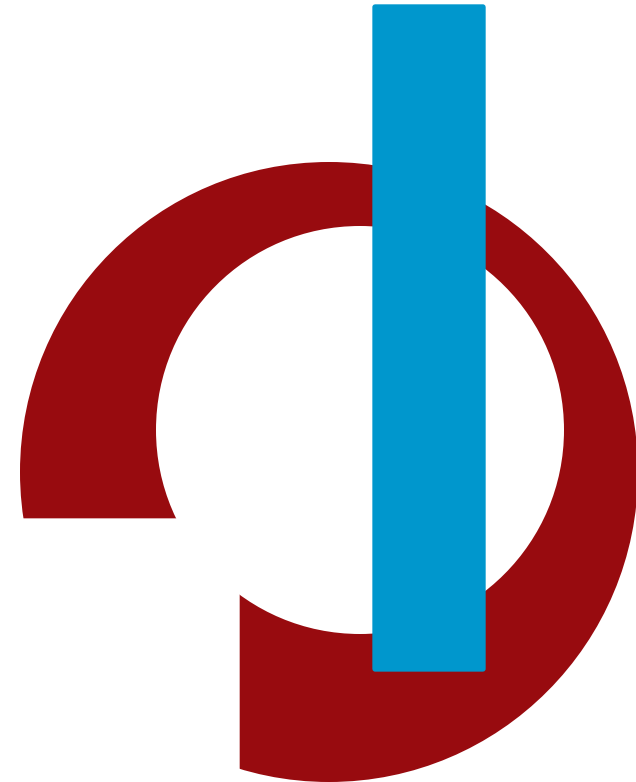
A large red circle representing a geometry with a priority of 5.1.

Vanadium
priority = 5.1

Geometry

Priority

- Each geometry is given a priority
- When two geometries overlap, the overlapping region has the material of the highest priority
- This allows the user to create complex geometries

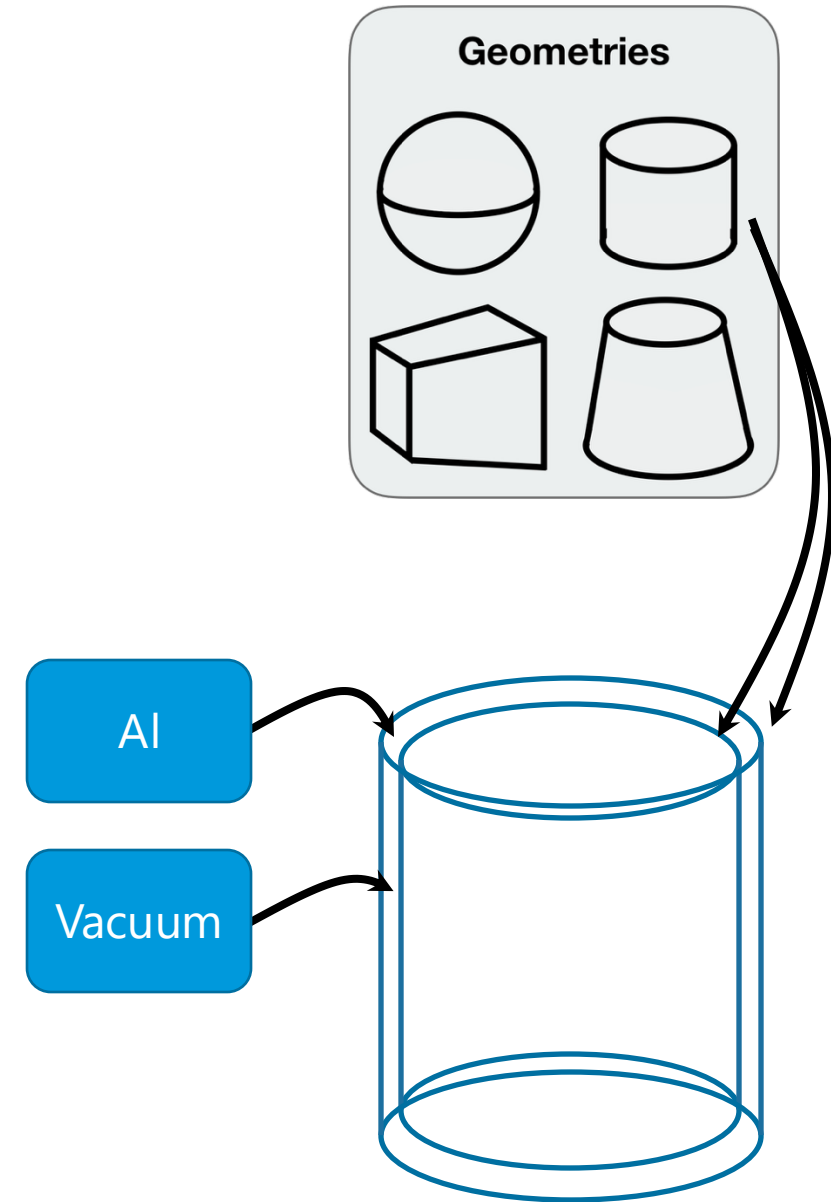


Geometry

Modular geometry

COMPONENT cryostat_shell = Union_cylinder(
radius = 0.15, height = 0.4,
priority = 10, material_string = "Al")
AT (0,0,0) RELATIVE target
ROTATED (0,0,0) RELATIVE target

COMPONENT cryostat_vacuum = Union_cylinder(
radius = 0.147, height = 0.39,
priority = 11, material_string = "Vacuum")
AT (0,0,0) RELATIVE target
ROTATED (0,0,0) RELATIVE target



Union master

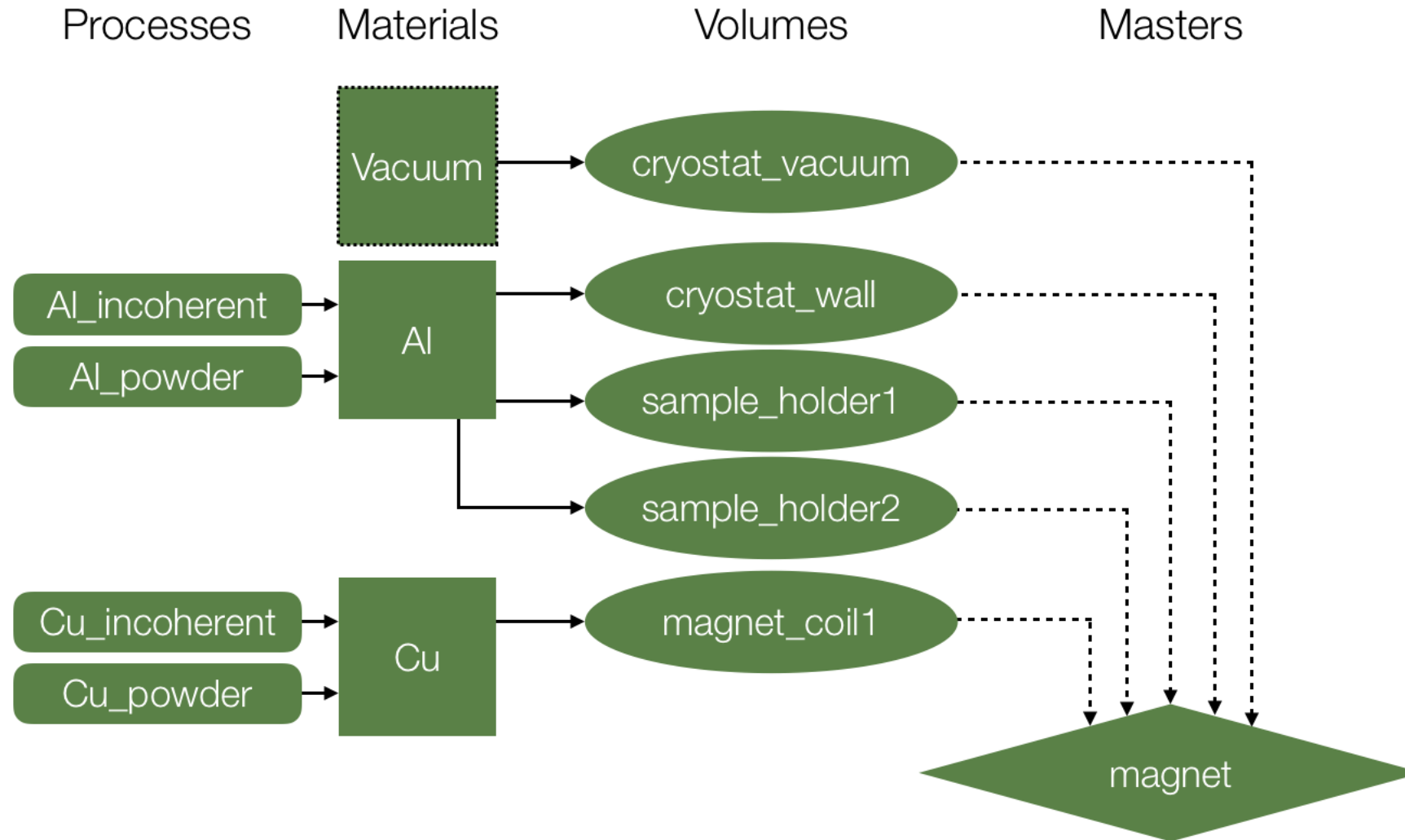
Where the simulation actually happens

- All components discussed doesn't actually perform any simulation
- Process and geometry components describe the system to be simulated
- The **Union_master** component performs the simulation
- A Union_master must be inserted, and will simulate all Union geometry components defined so far
- No inputs are necessary, as all the information is already in the processes and geometry

```
COMPONENT master = Union_master()  
AT (0,0,0) ABSOLUTE
```

Union master

Overview of a Union setup



Introduction

How does this improve the situation?

- Sample holders with complicated geometry
- Consists of many different materials
- No clear order in which neutrons would pass through
- Could have co aligned crystals
- Twinned crystals
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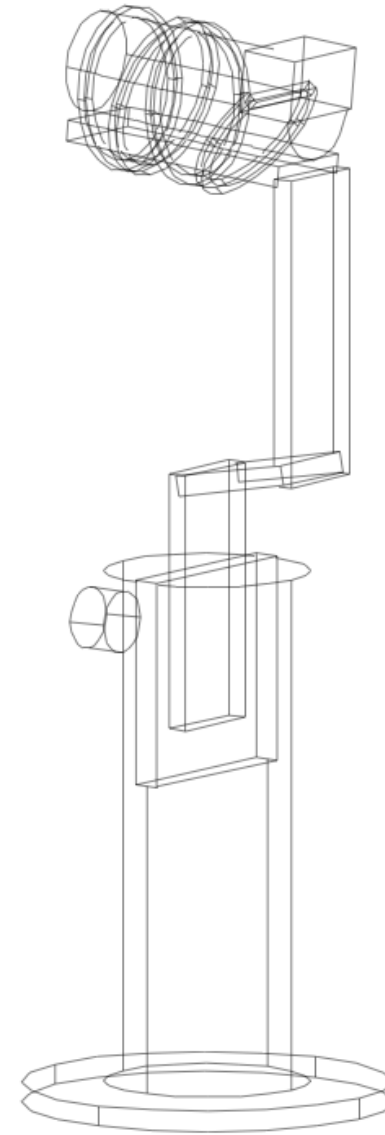


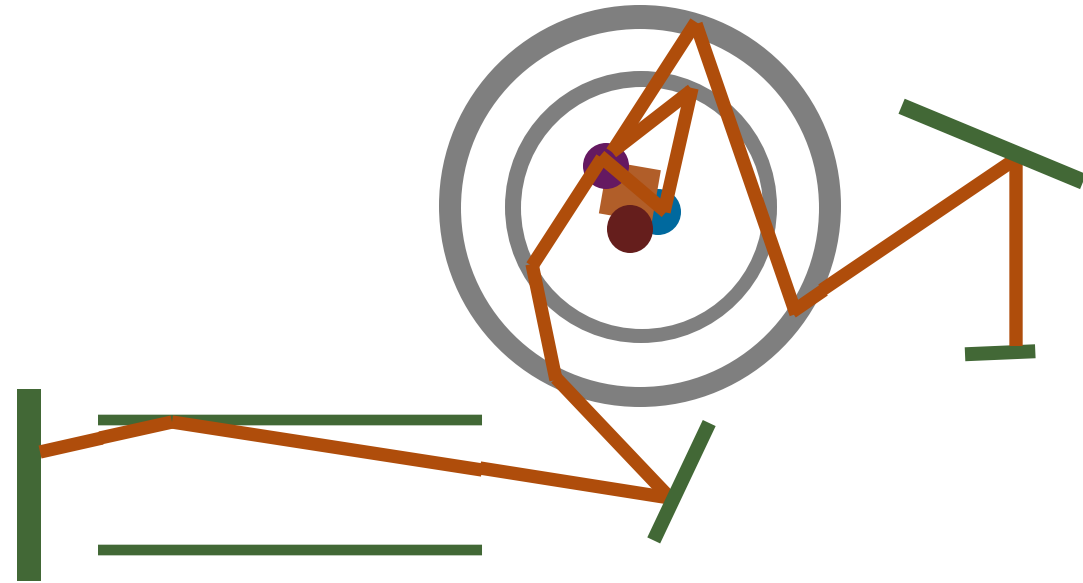
Photo by Pia Ray Jensen



Multiple scattering

Logic in McStas instrument

- Multiple scattering handled in Union_master component
- Neutron handed back to instrument component sequence



Simulated 3D space

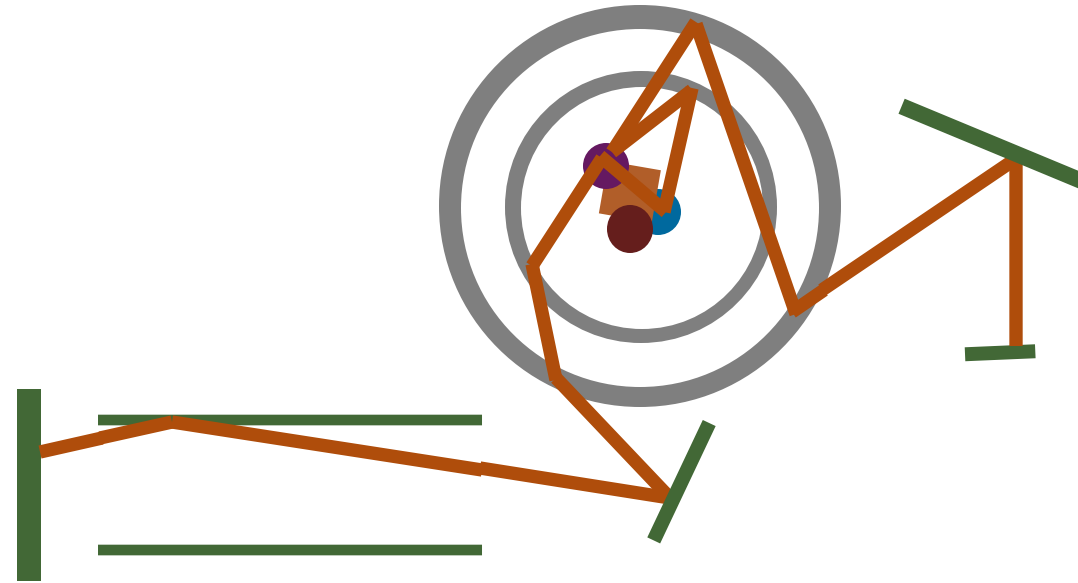
Instrument file



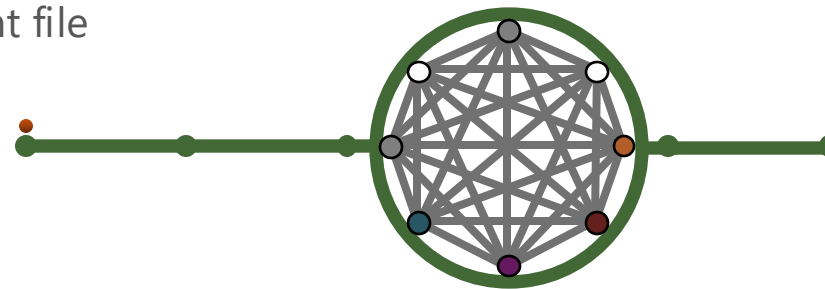
Multiple scattering

Logic in McStas instrument

- Multiple scattering handled in Union_master component
- Neutron handed back to instrument component sequence
- Ray can go to geometries in any order



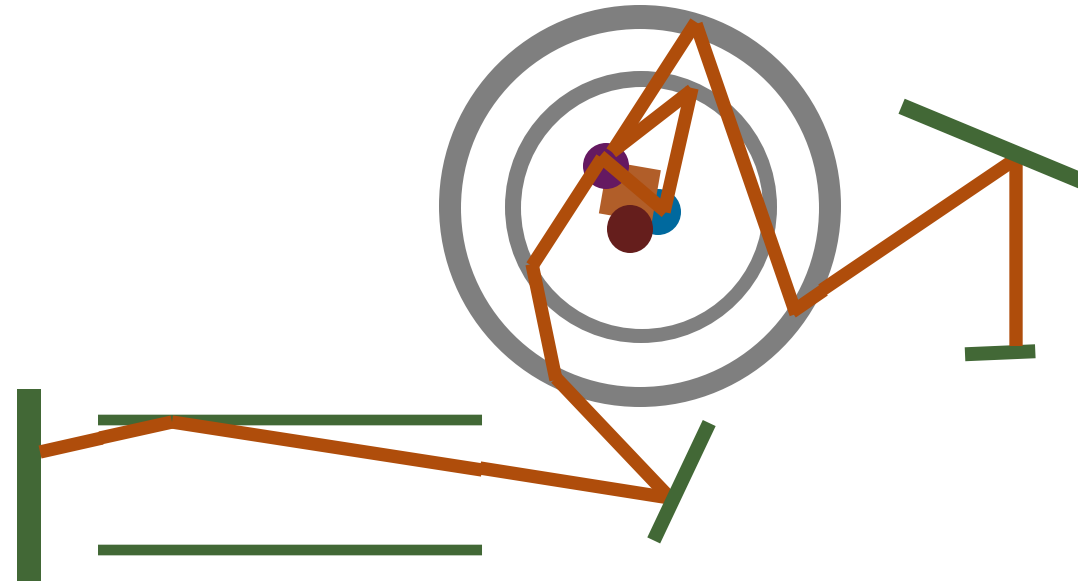
Simulated 3D space
Instrument file



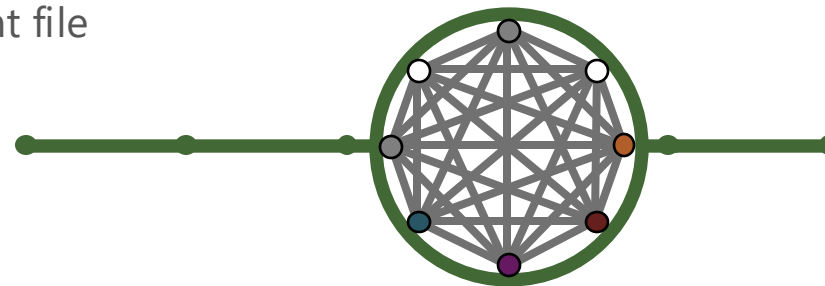
Multiple scattering

Logic in McStas instrument

- Multiple scattering handled in Union_master component
- Neutron handed back to instrument component sequence
- Ray can go to geometries in any order
- Internal logic network reduce number of intersection calculations, improving execution speed

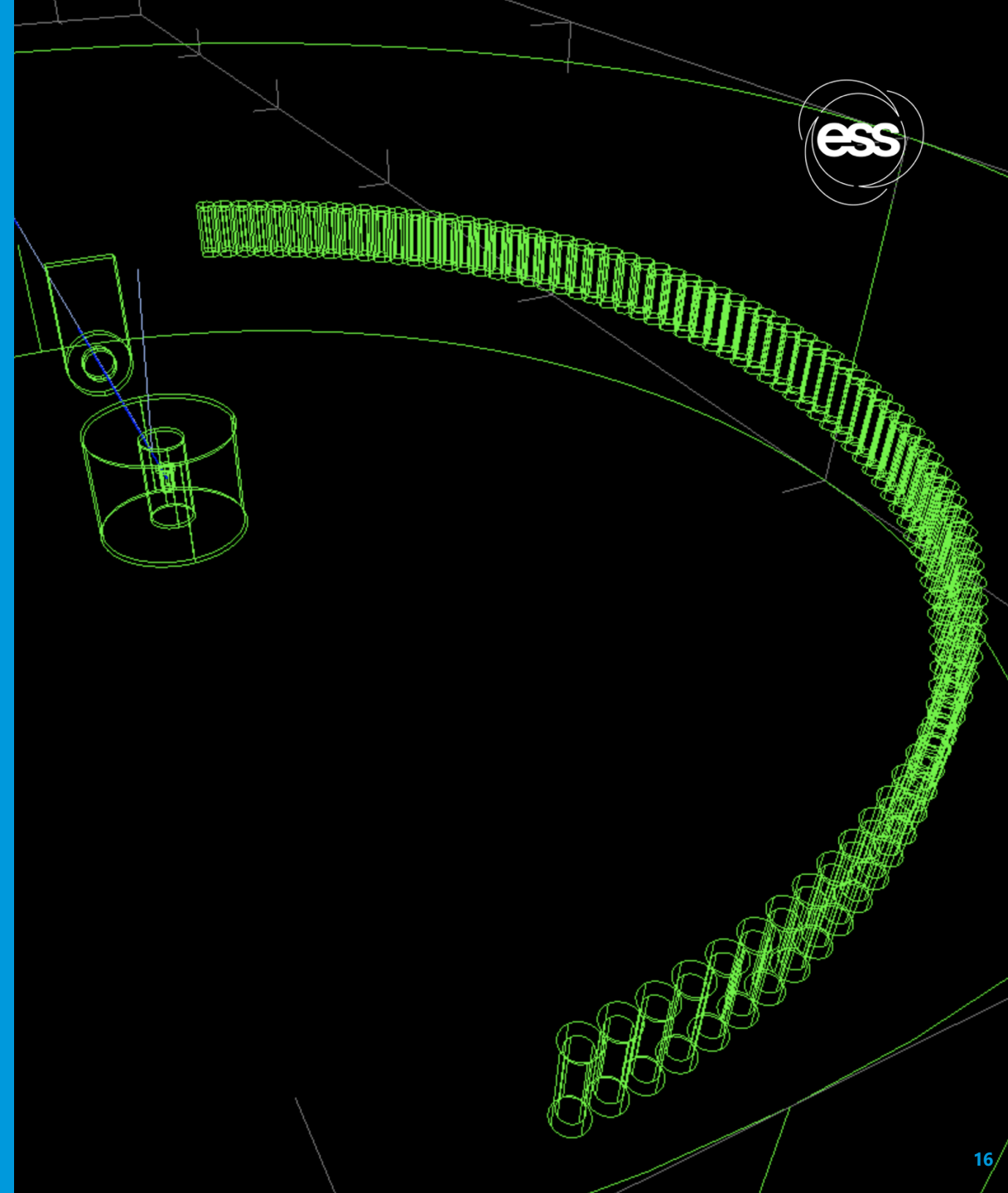


Simulated 3D space
Instrument file



2

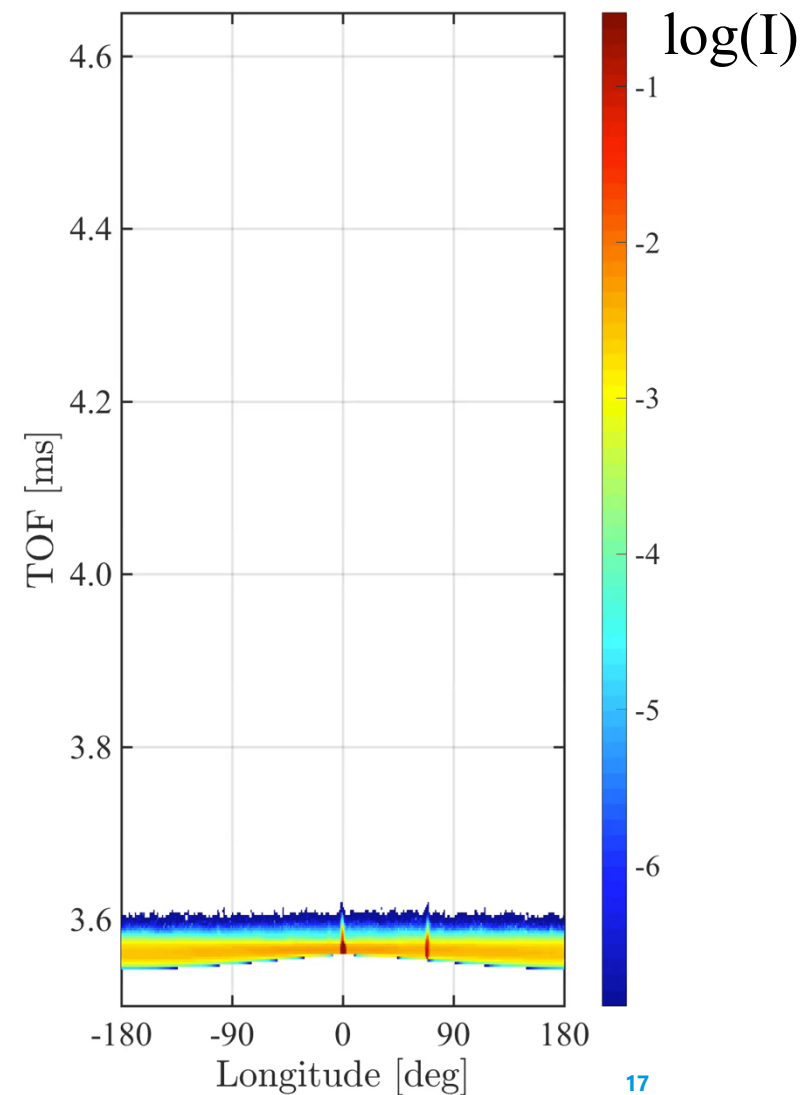
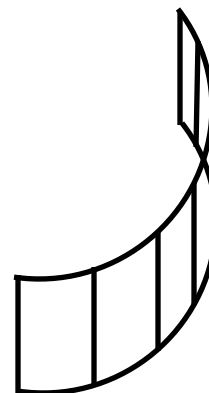
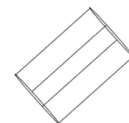
Demonstration



Demonstration

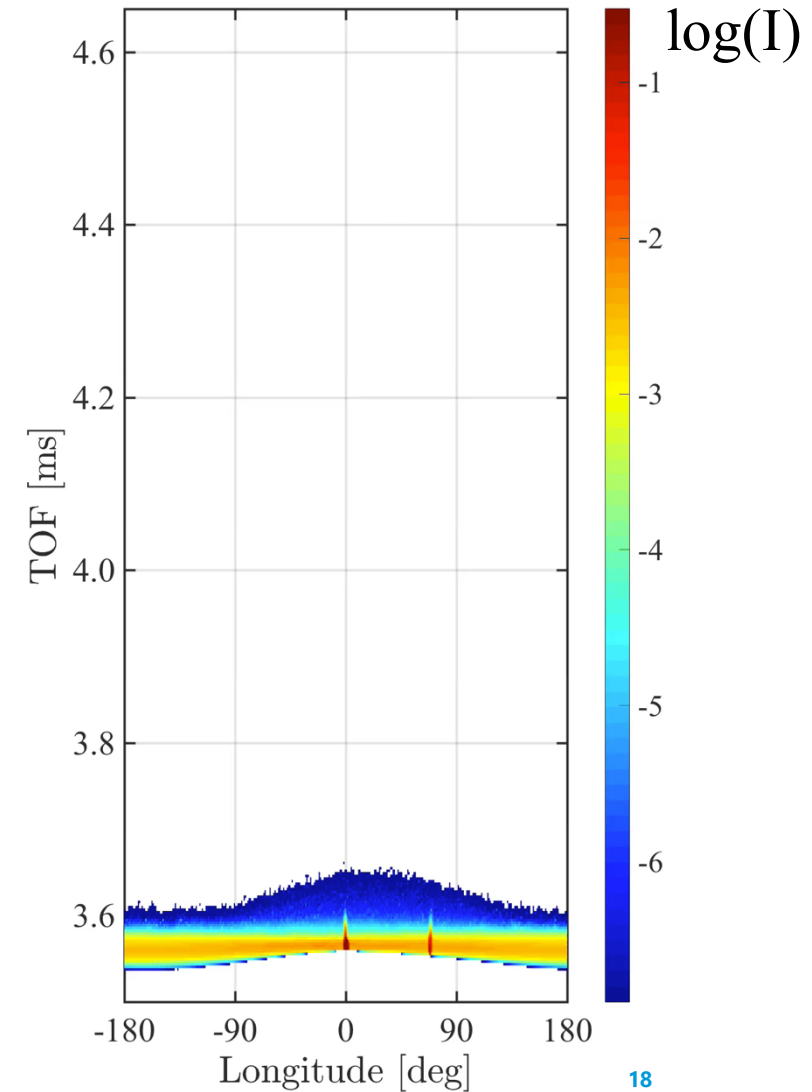
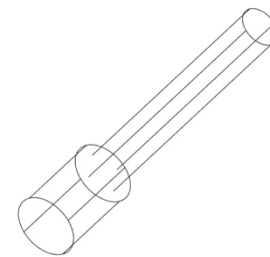


5 meV beam



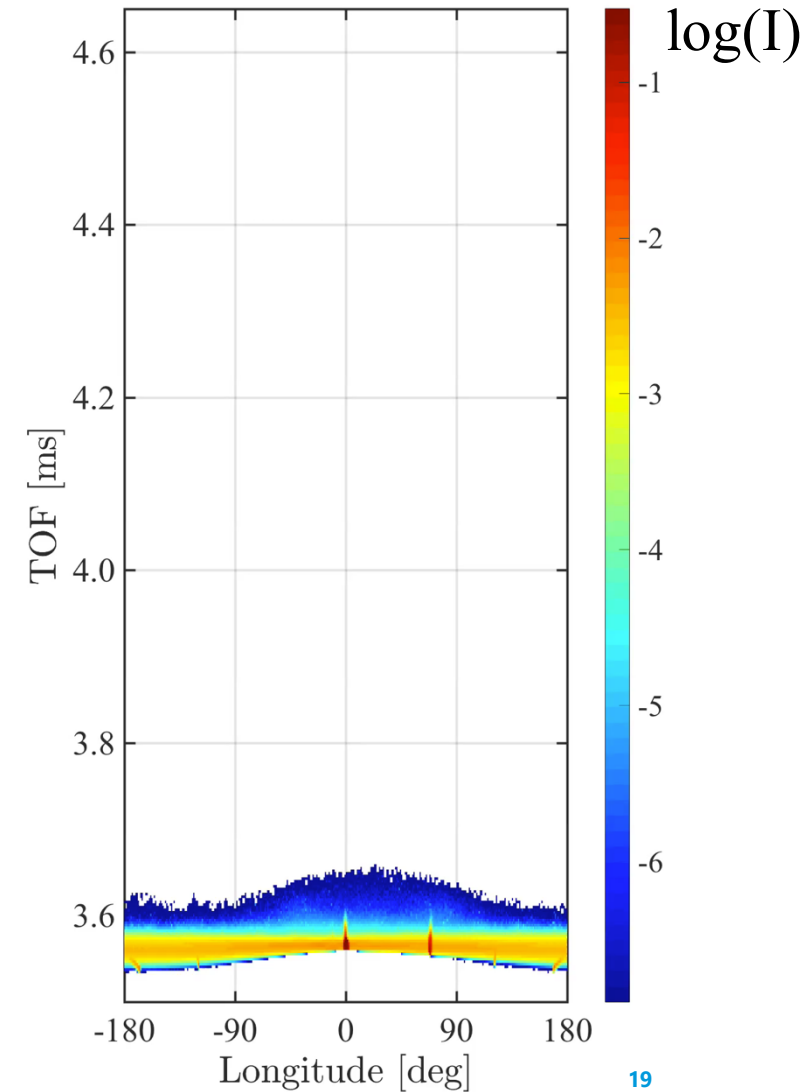
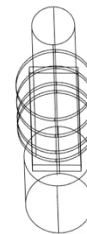
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Demonstration



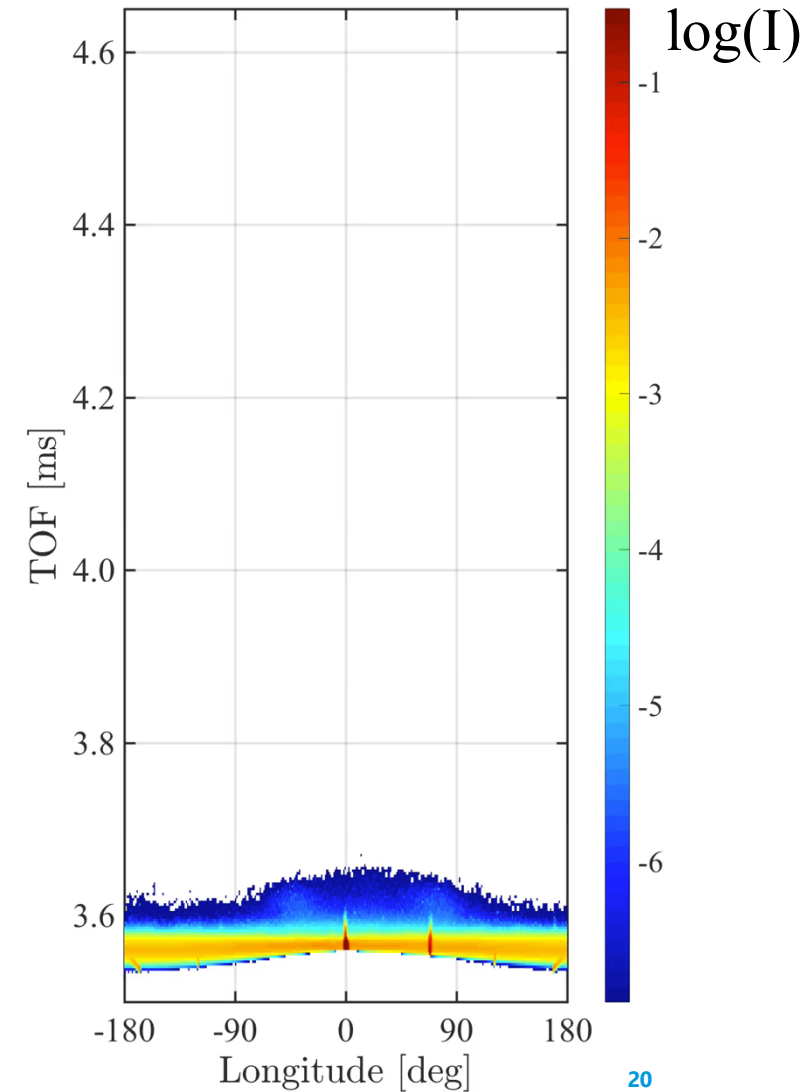
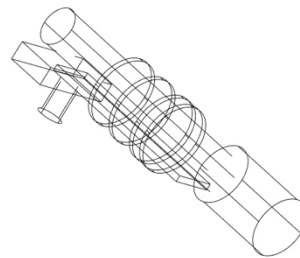
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Demonstration



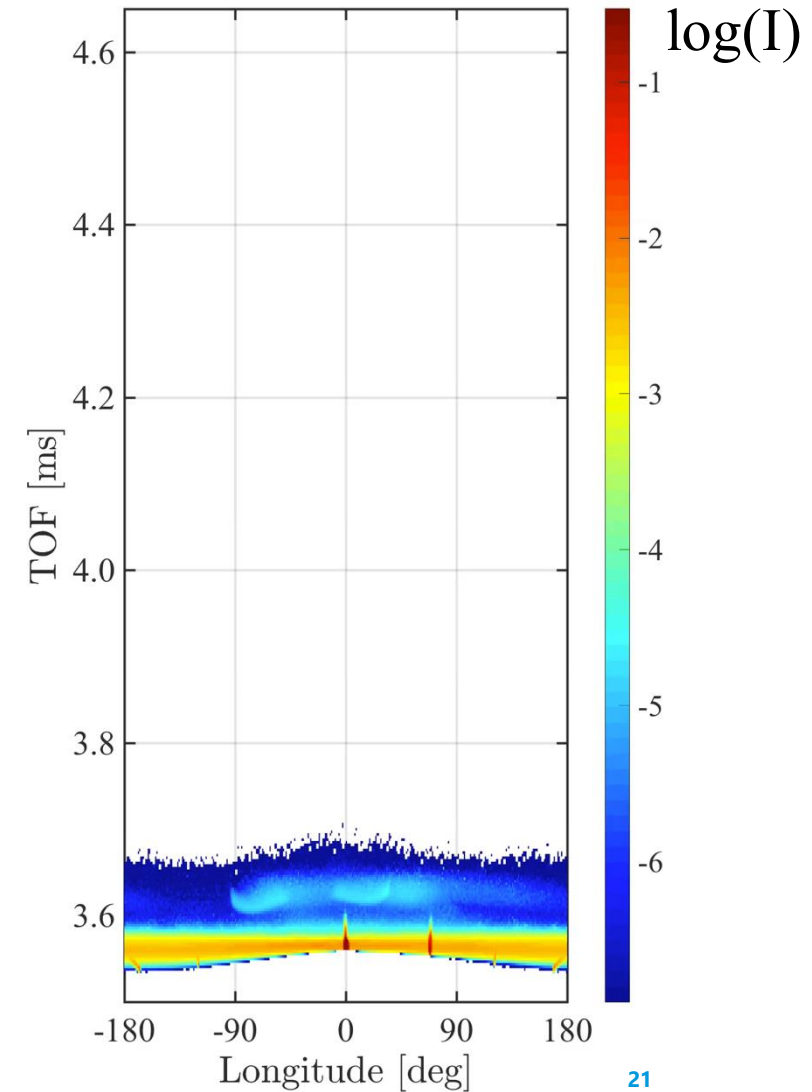
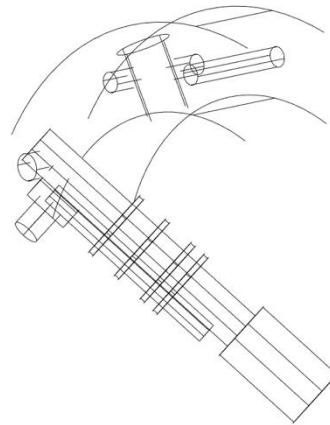
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Demonstration



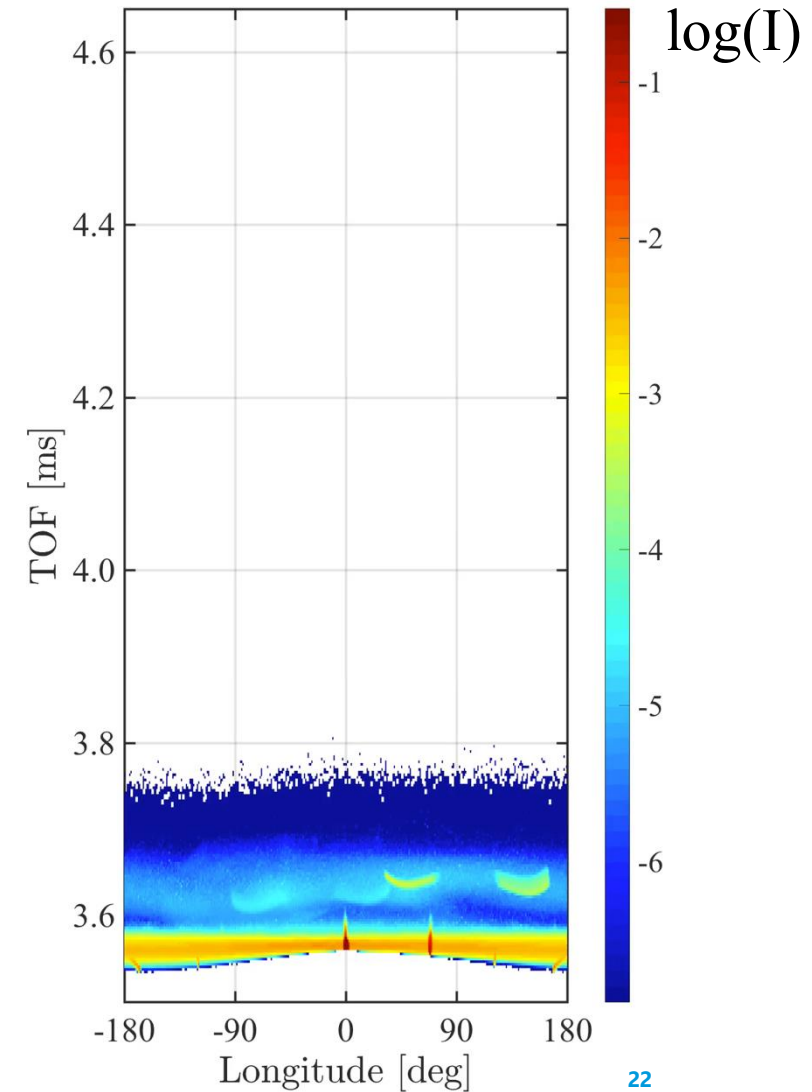
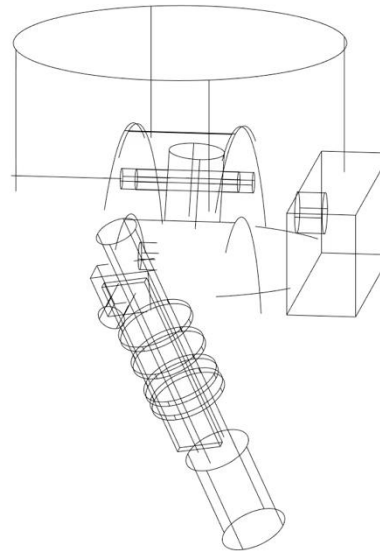
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Demonstration



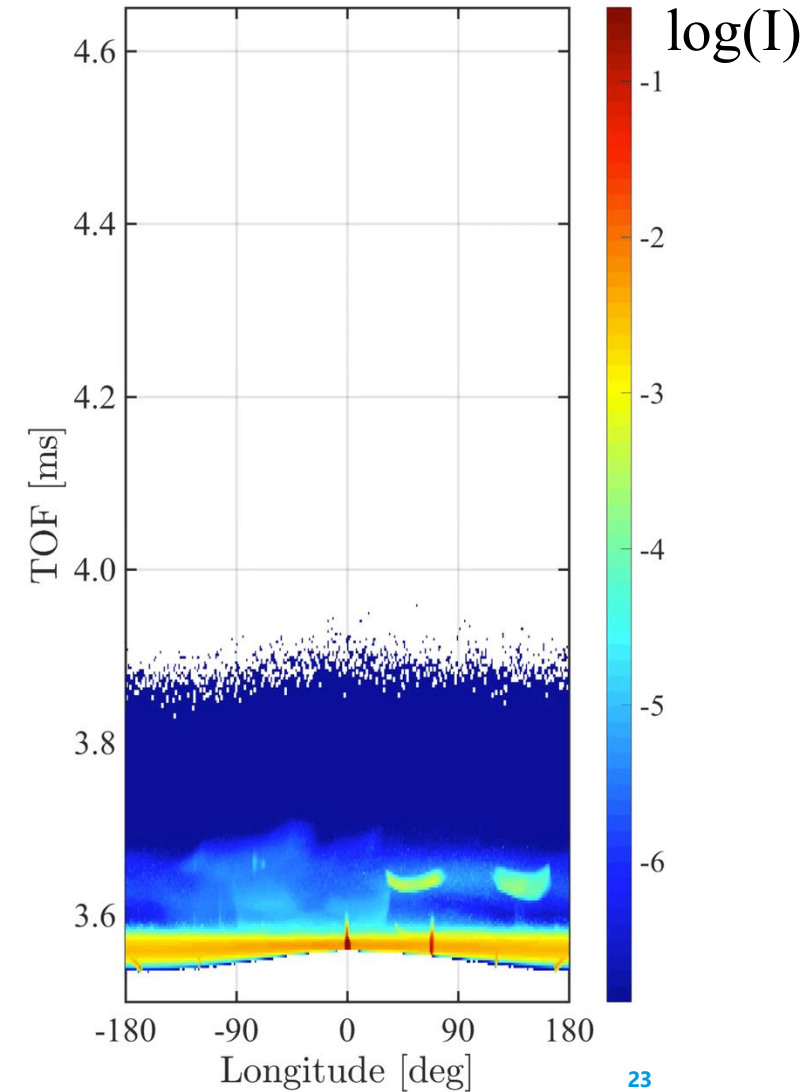
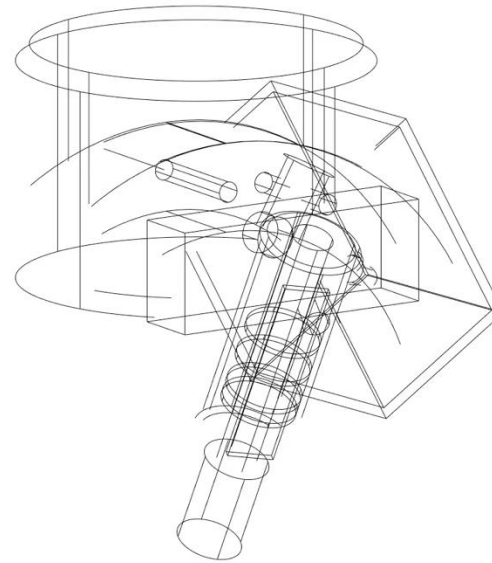
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Demonstration



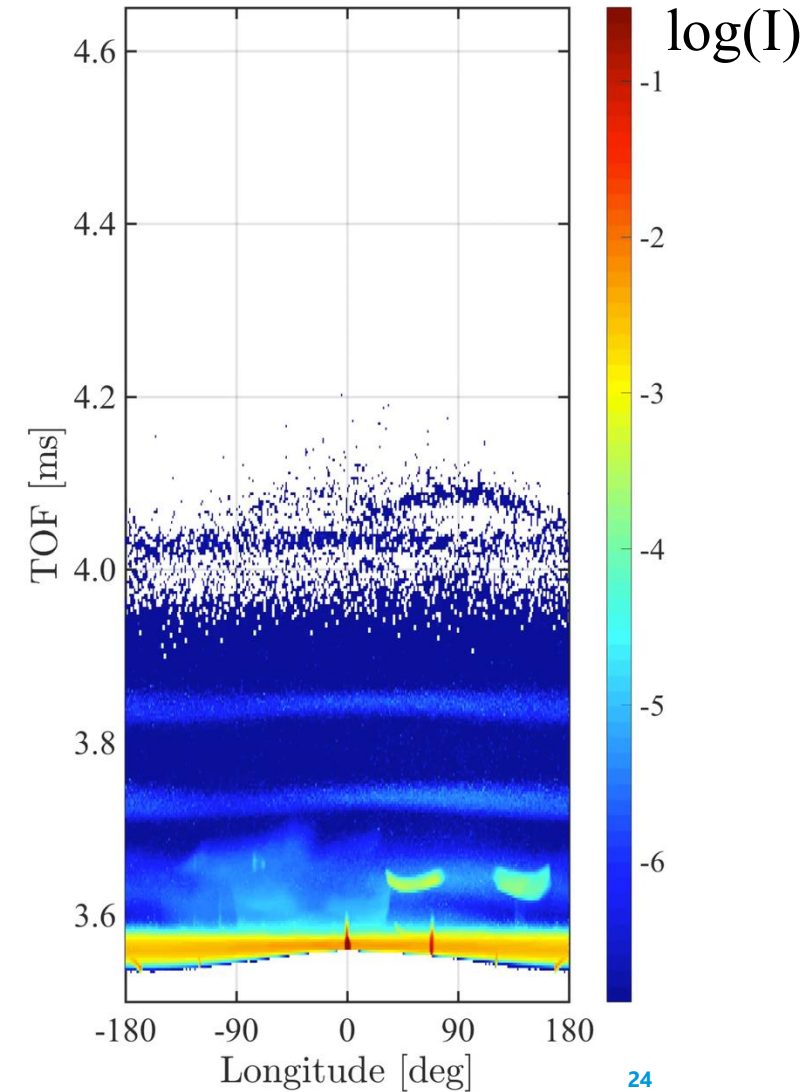
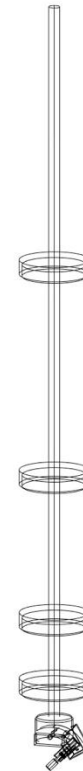
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Demonstration



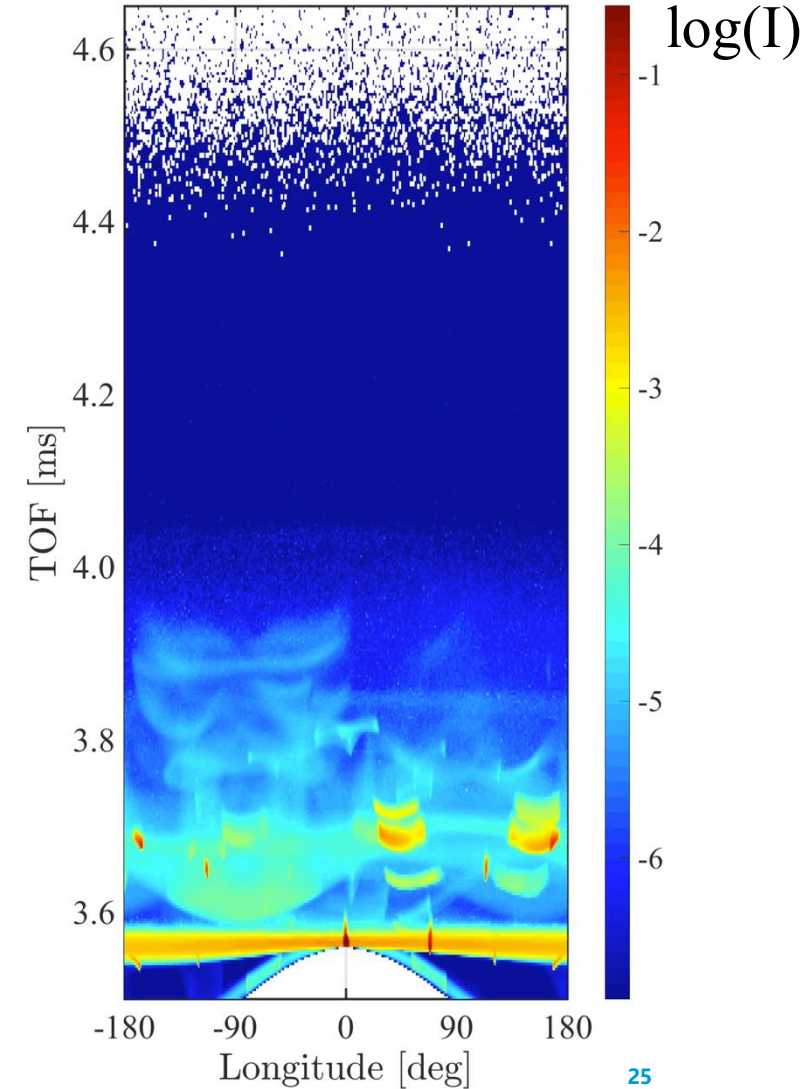
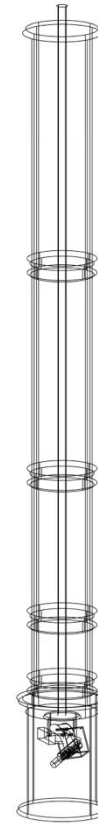
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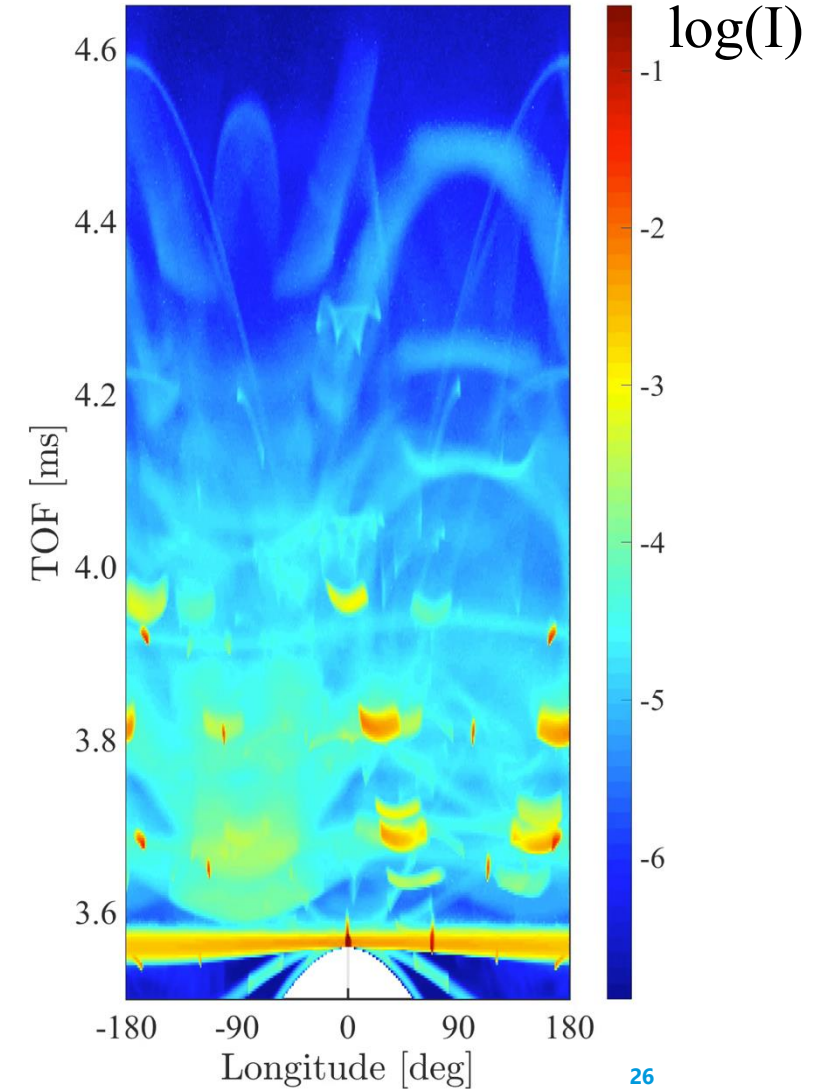
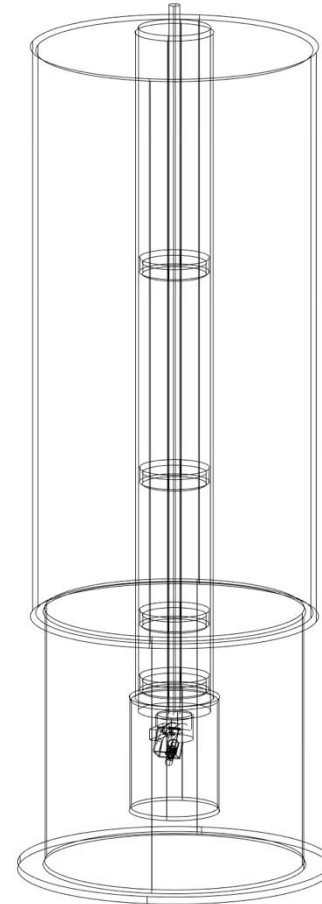
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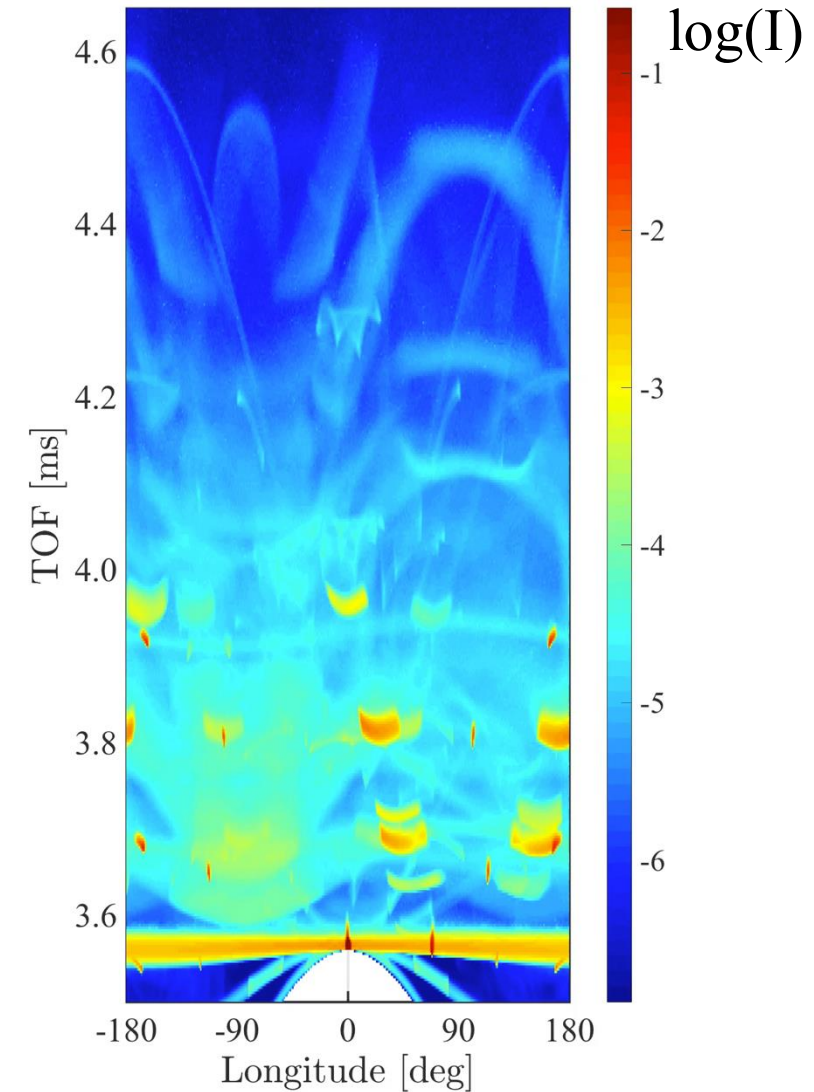
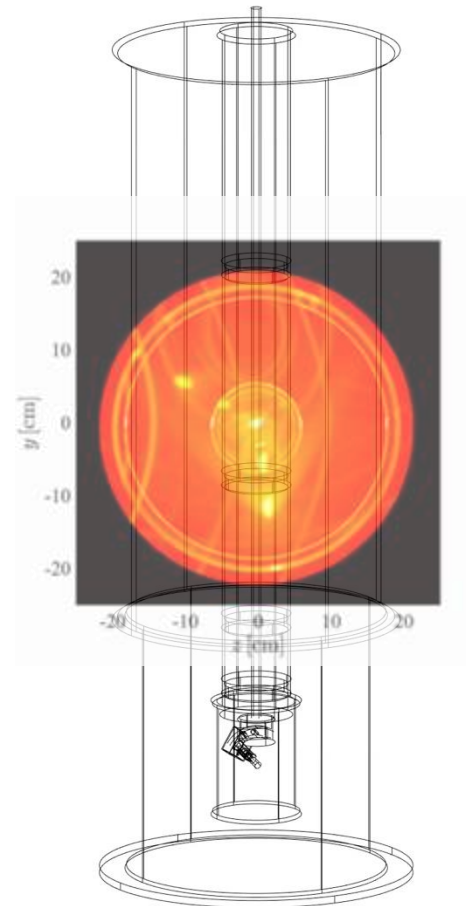
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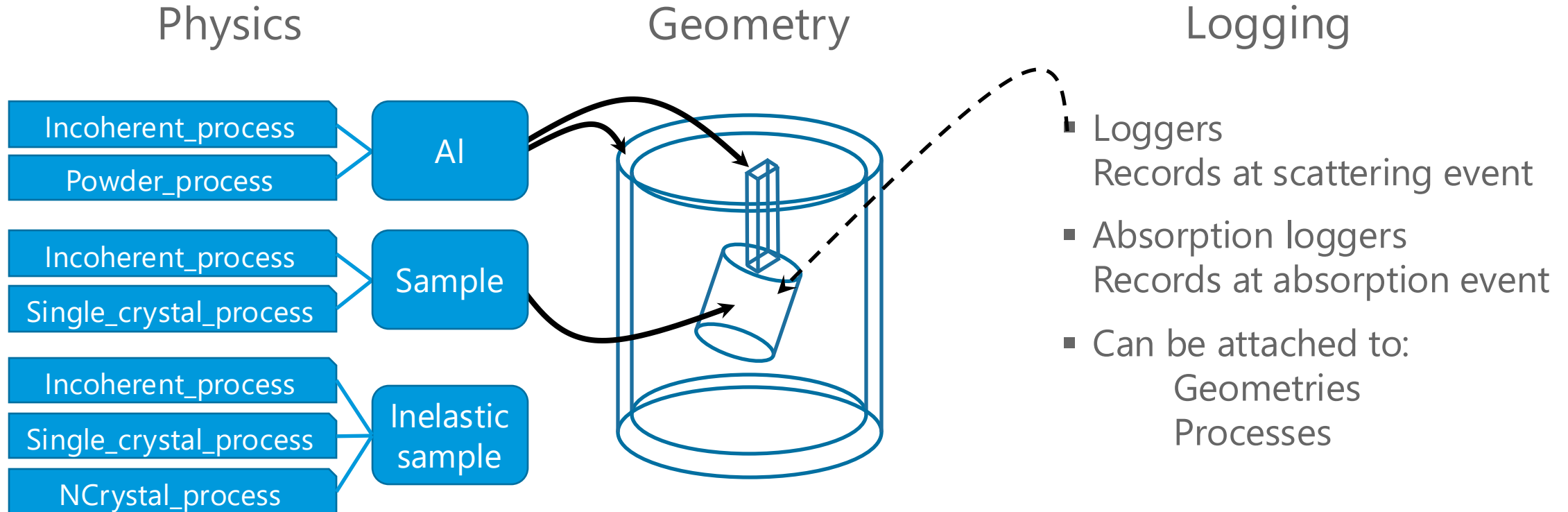


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Logger components

Recording information during simulation

- Loggers can be used to understand what is going on during simulation
- Can either log everything, or be attached to a geometry or process

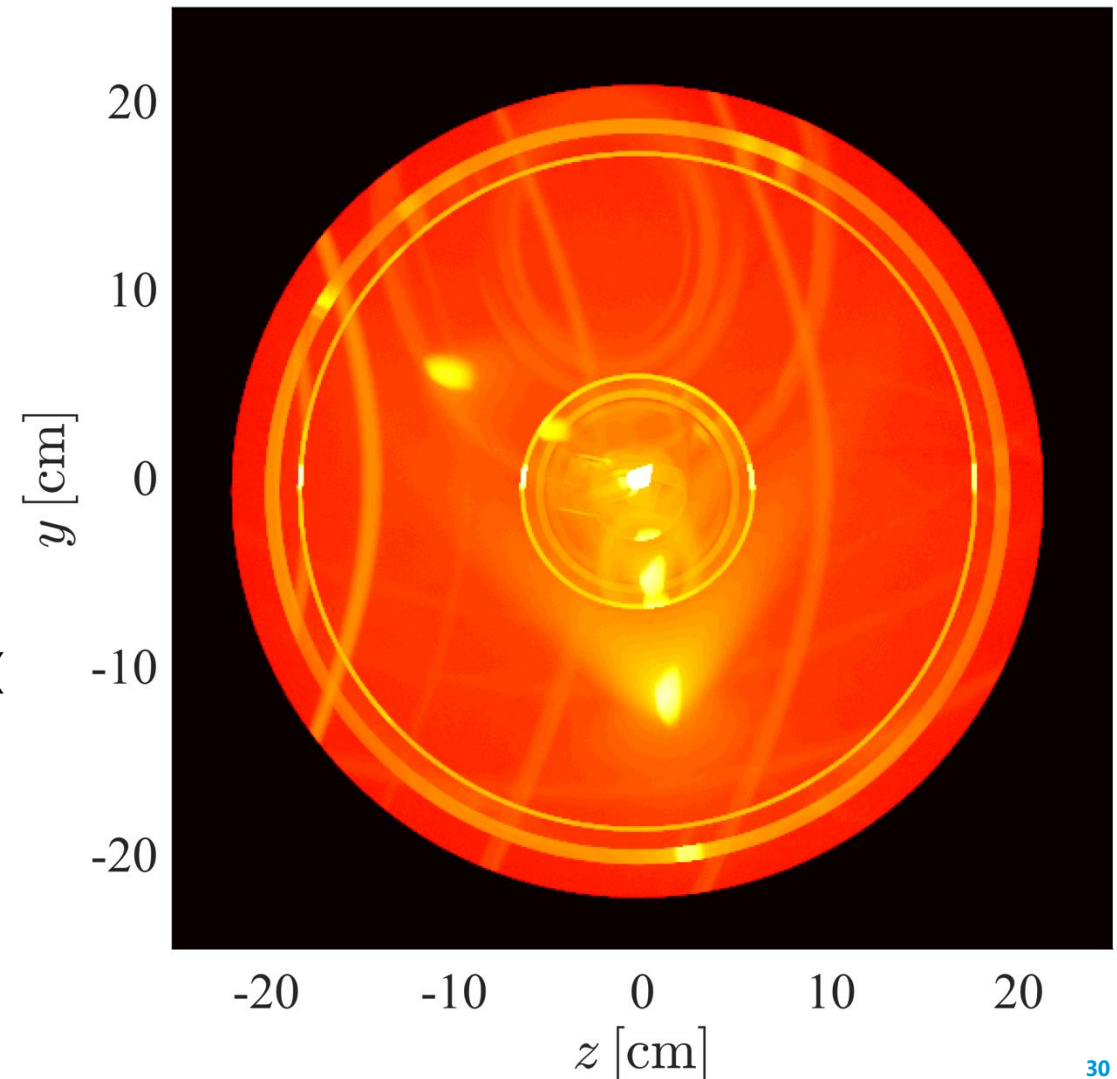


Logger components

Recording information during simulation

- Need to understand the simulation
- Record information during scattering
- Here spatial distribution of scattered intensity from above

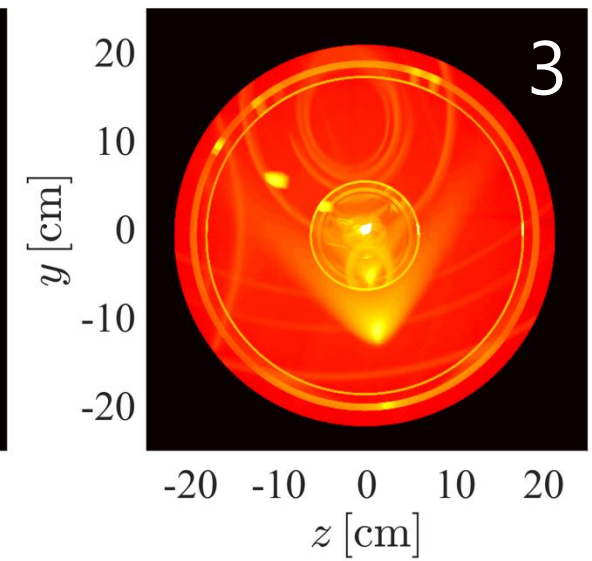
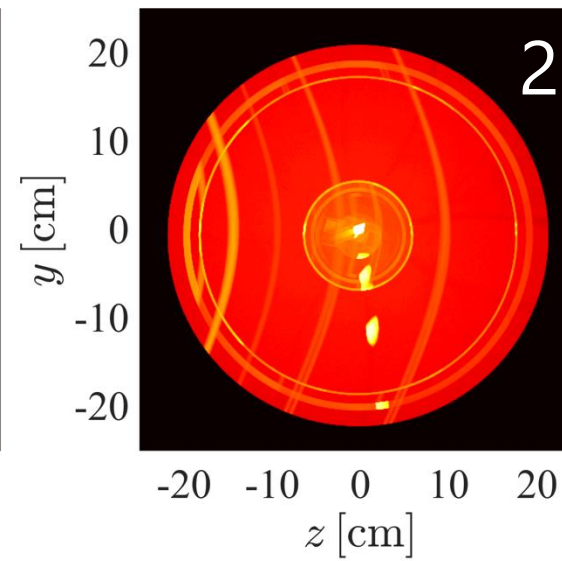
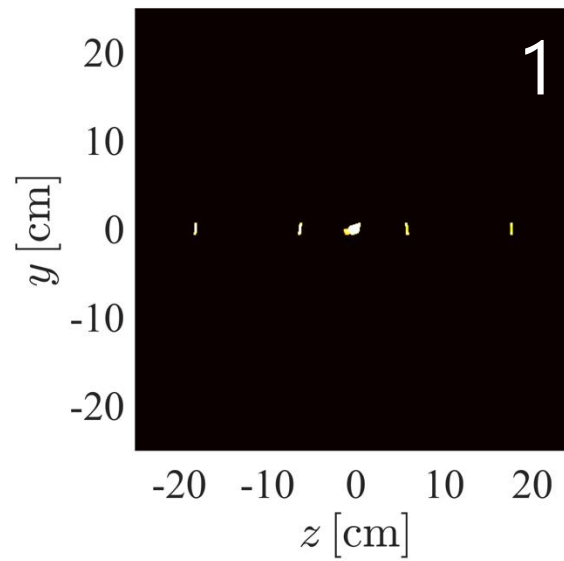
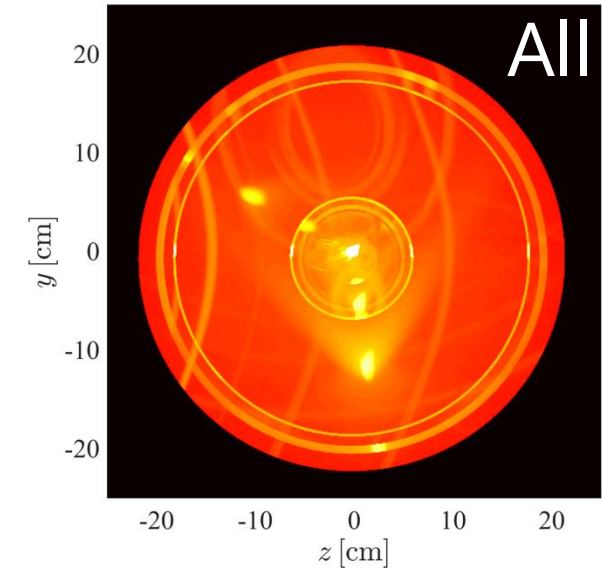
```
COMPONENT scattering_zx = Union_logger_2D_space(
  D_direction_1="z", D1_min=-0.15, D1_max=0.15,
  n1=300,
  D_direction_2="x", D2_min=-0.15, D2_max=0.15,
  n2=300,
  filename="scattering_zx.dat")
AT (0,0,0) RELATIVE sample_position
```



Logger components

Recording information during simulation

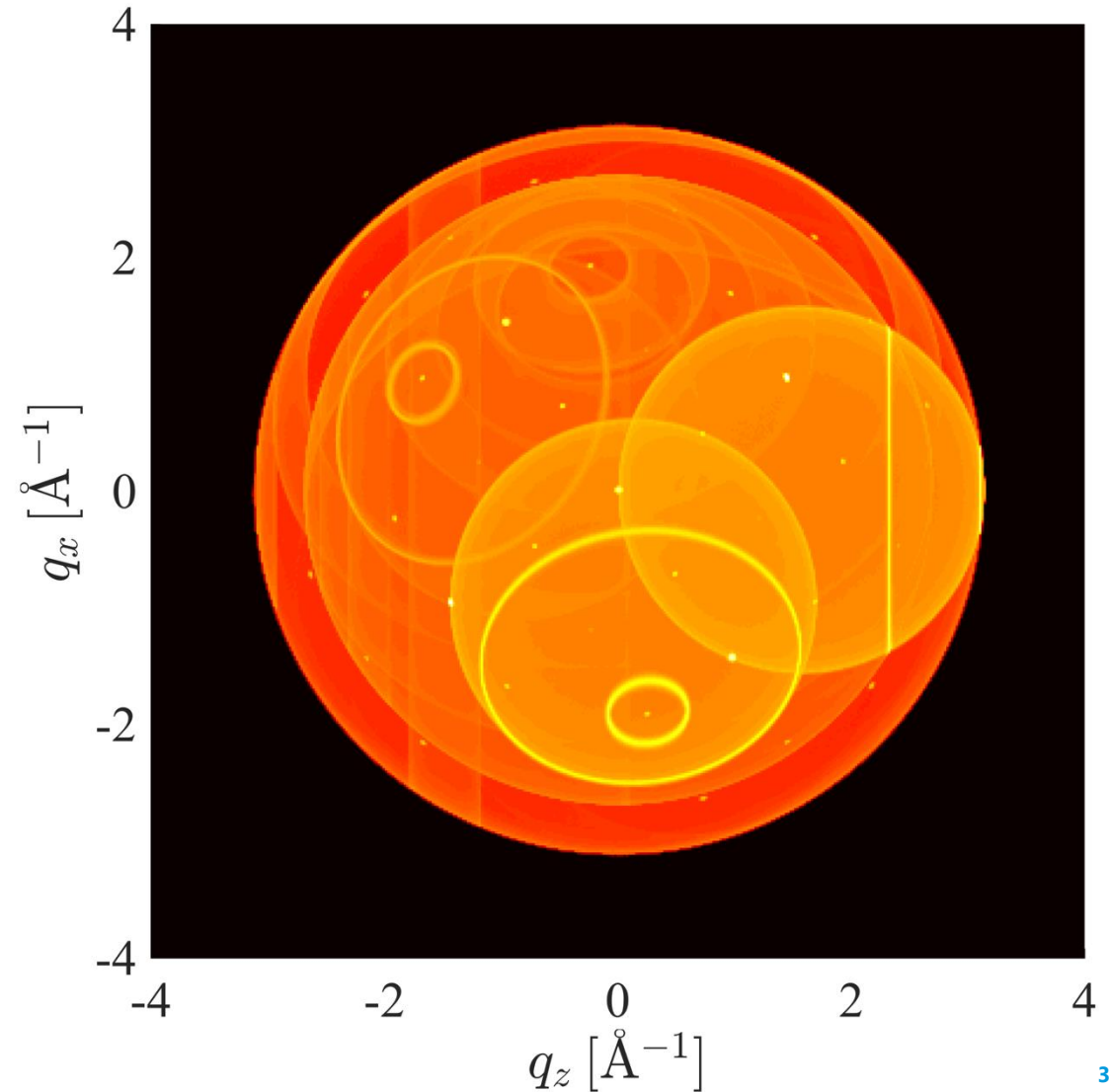
- A specific scattering order can be shown
- Improves understanding of what occurred



Logger components

Recording information during simulation

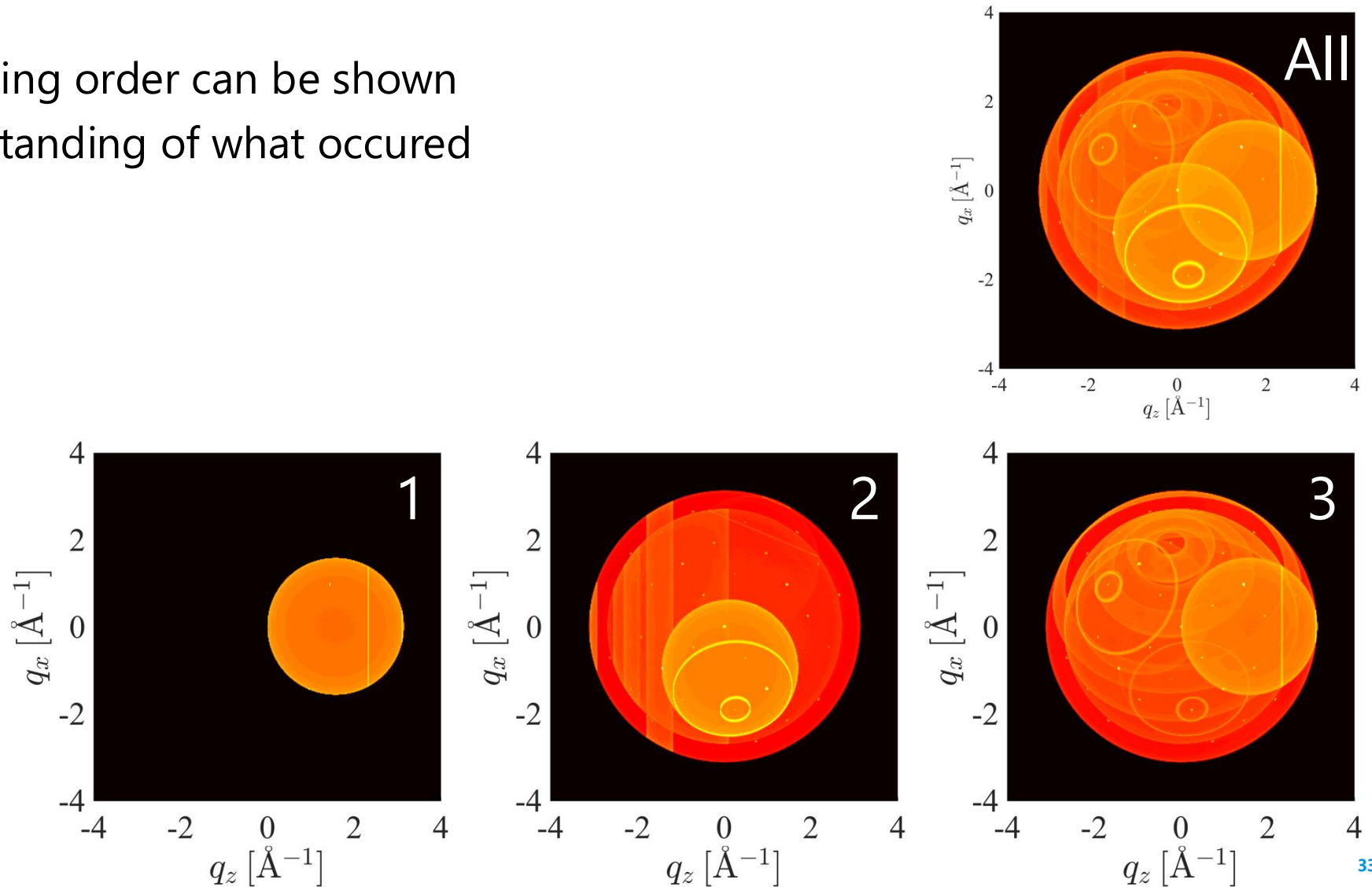
- Here the scattering vector is recorded
- Visible Bragg peaks and Debye Scherrer cones

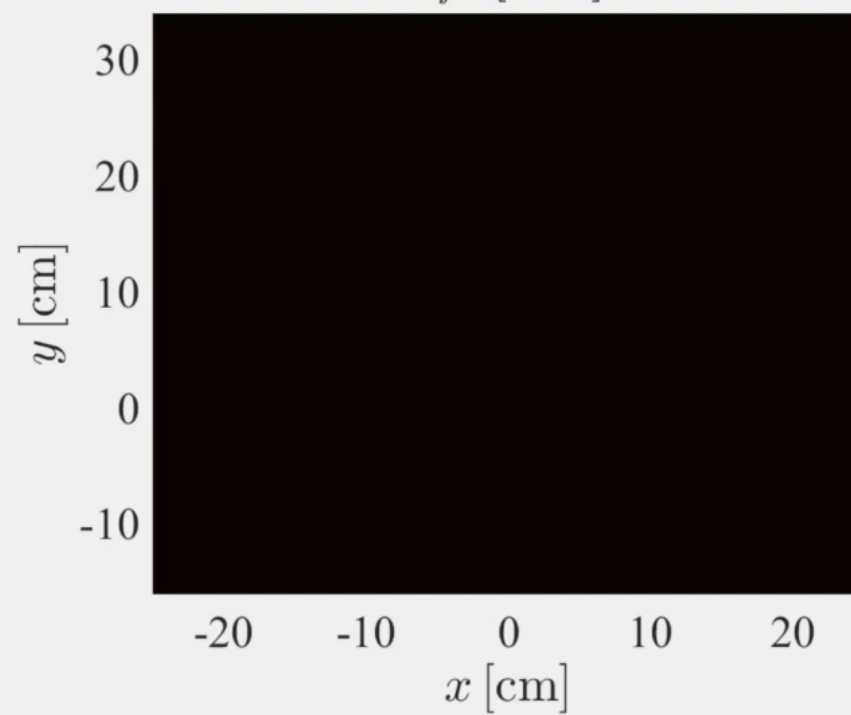
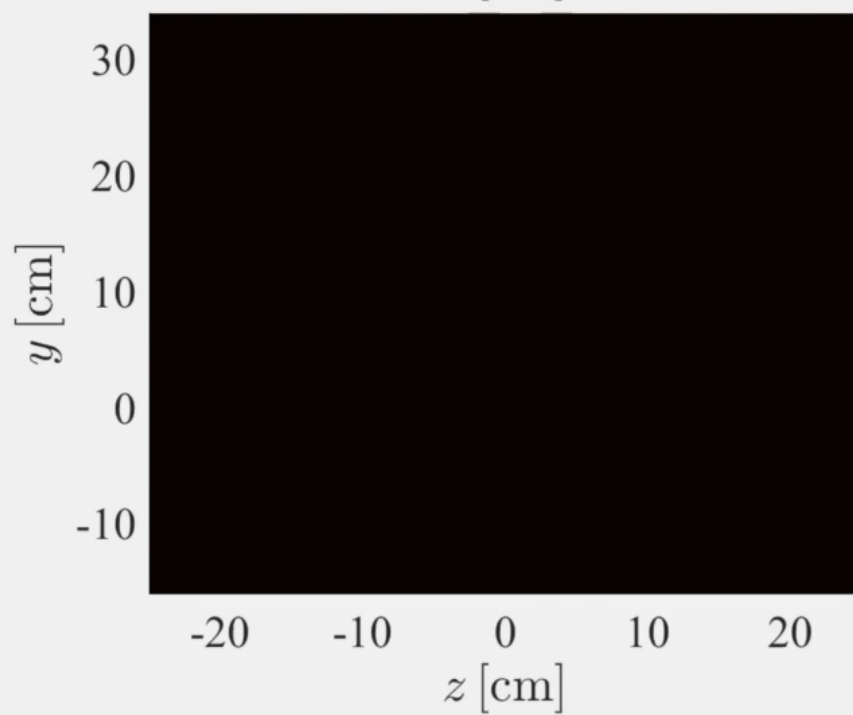
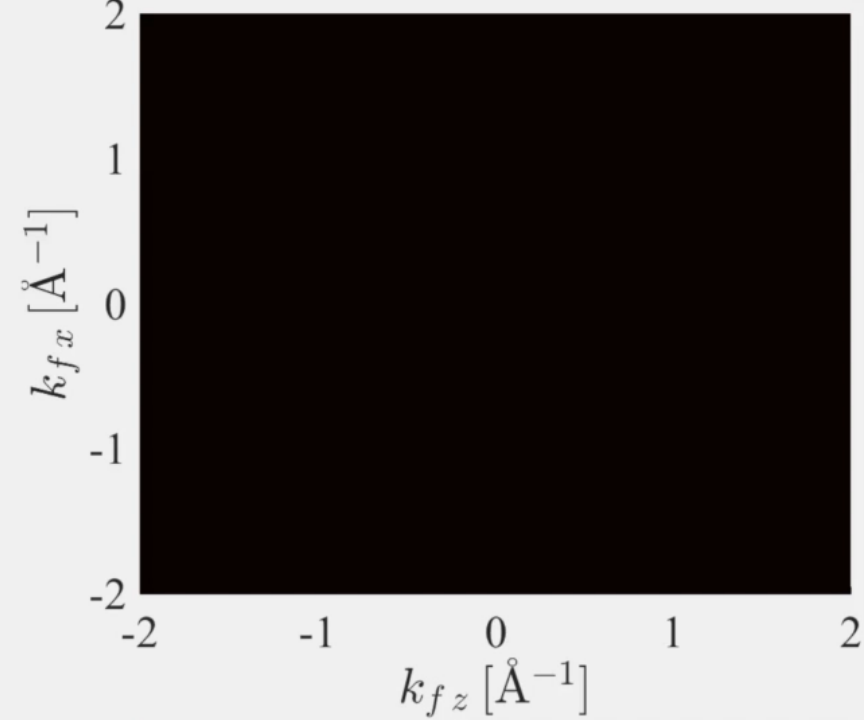
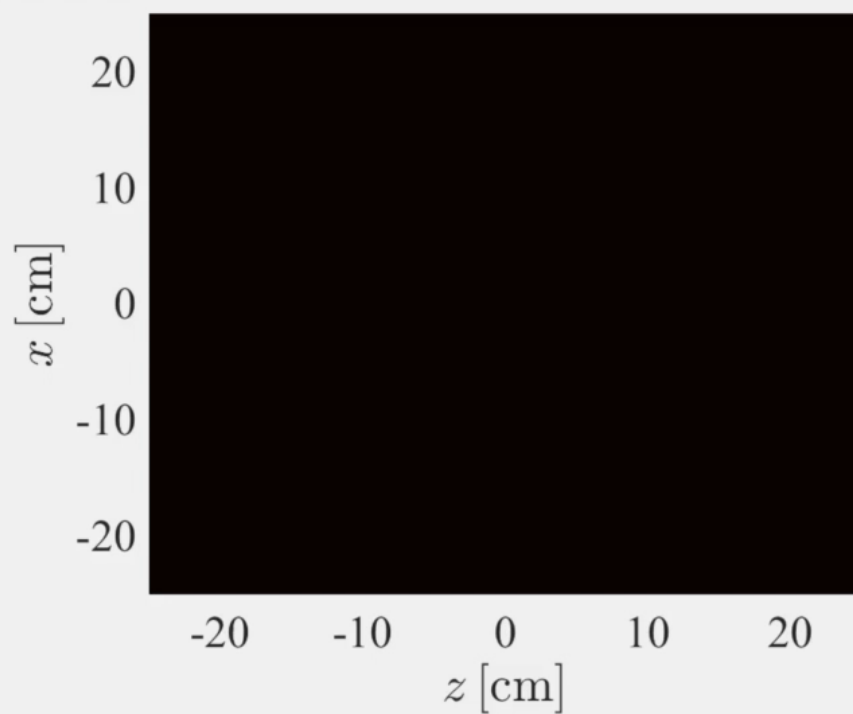


Logger components

Recording information during simulation

- A specific scattering order can be shown
- Improves understanding of what occurred



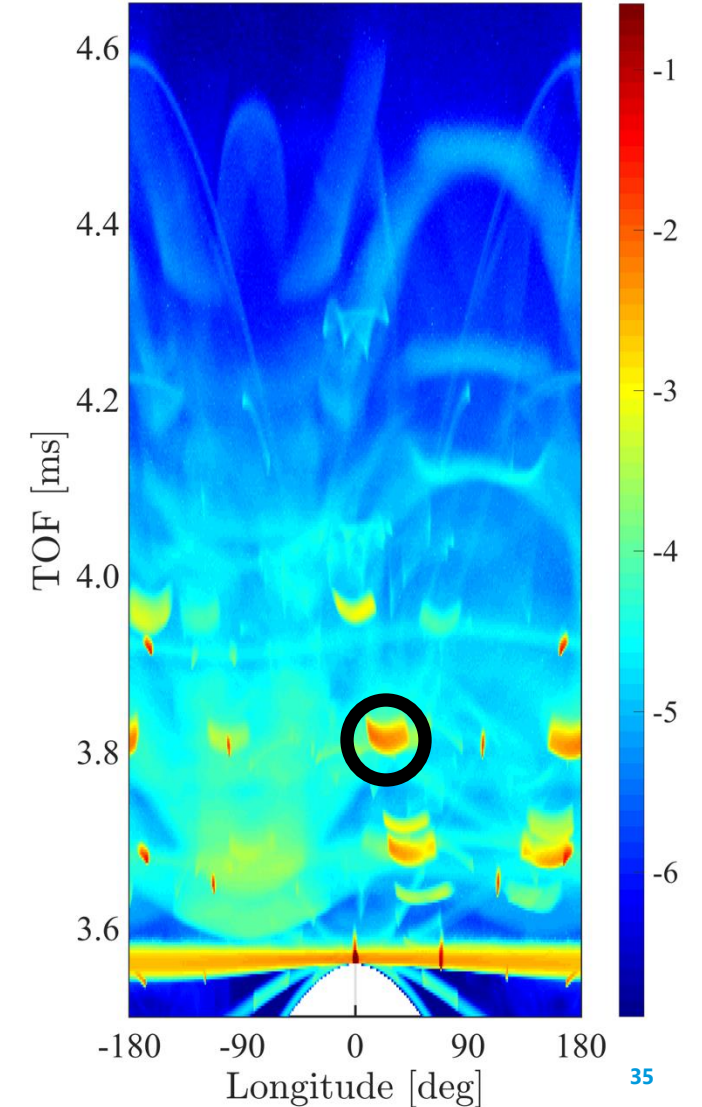
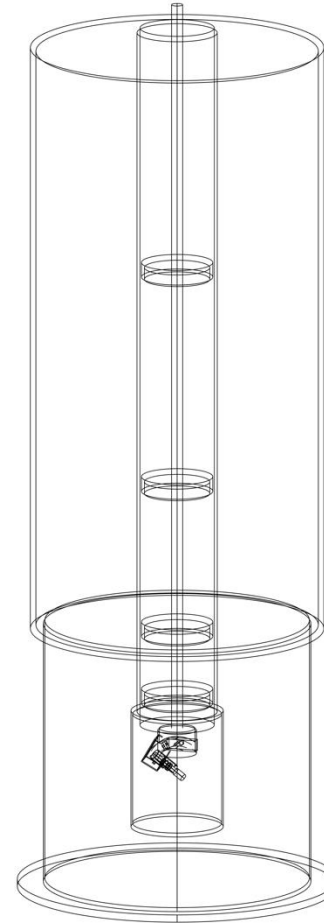


Conditional components

Limiting logger components to record interesting neutrons



- Wish to understand how individual background features arose



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Conditional components

Limiting logger components to record interesting neutrons

- Conditional components set up some condition upon the final neutron, such as if it hits a certain area
- Conditional components modify existing loggers with *target_loggers*
- Can specify many loggers with comma separated list
- Current conditionals
 - Union_conditional_PSD
 - Union_conditional_standard

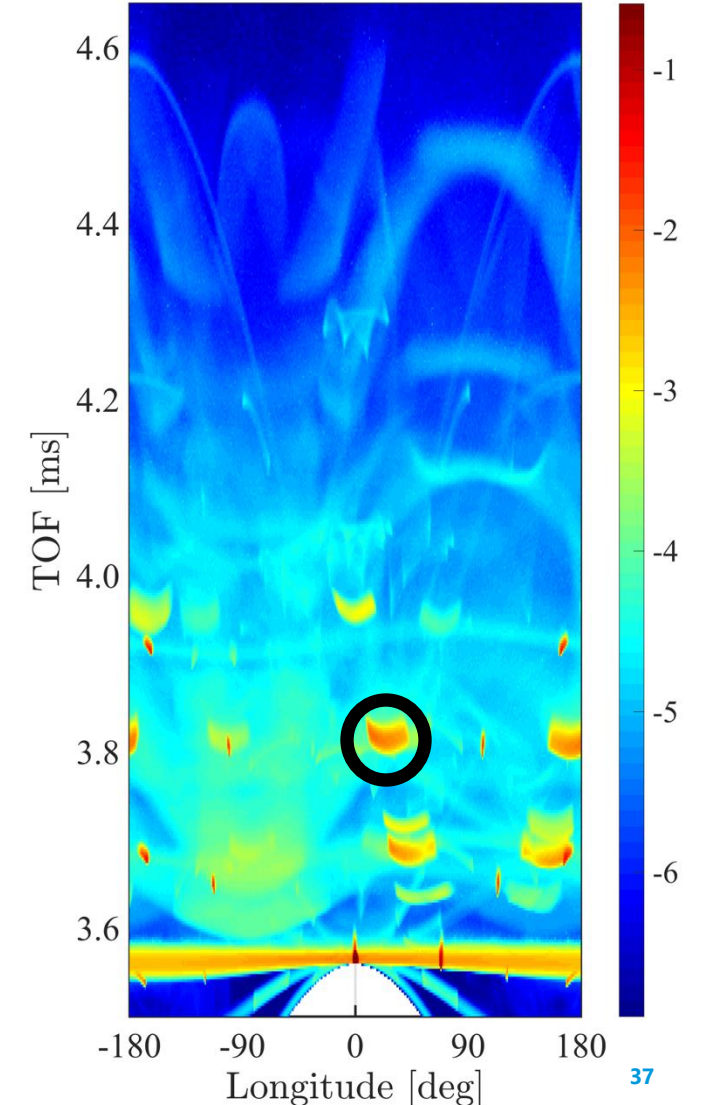
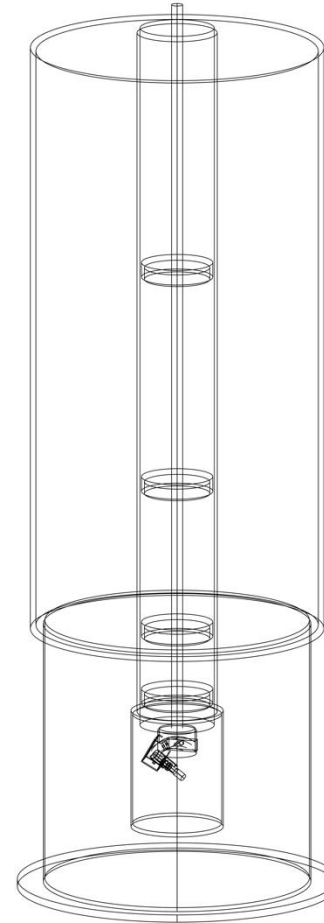
```
COMPONENT scattering_zx = Union_logger_2D_space(  
    D_direction_1 = "z", D1_min = -0.15, D1_max = 0.15, n1 =  
    300,  
    D_direction_2 = "x", D2_min = -0.15, D2_max = 0.15, n2 =  
    300,  
    filename = "scattering_zx.dat")  
AT (0,0,0) RELATIVE sample_position
```

```
COMPONENT conditional_psd = Union_conditional_PSD(  
    xwidth = 0.02, yheight = 0.02,  
    time_min = 2E-6, time_max = 8E-6  
    target_loggers = "scattering_zx")  
AT (0,0,1) RELATIVE sample_position
```

Conditional components

Limiting logger components to record interesting neutrons

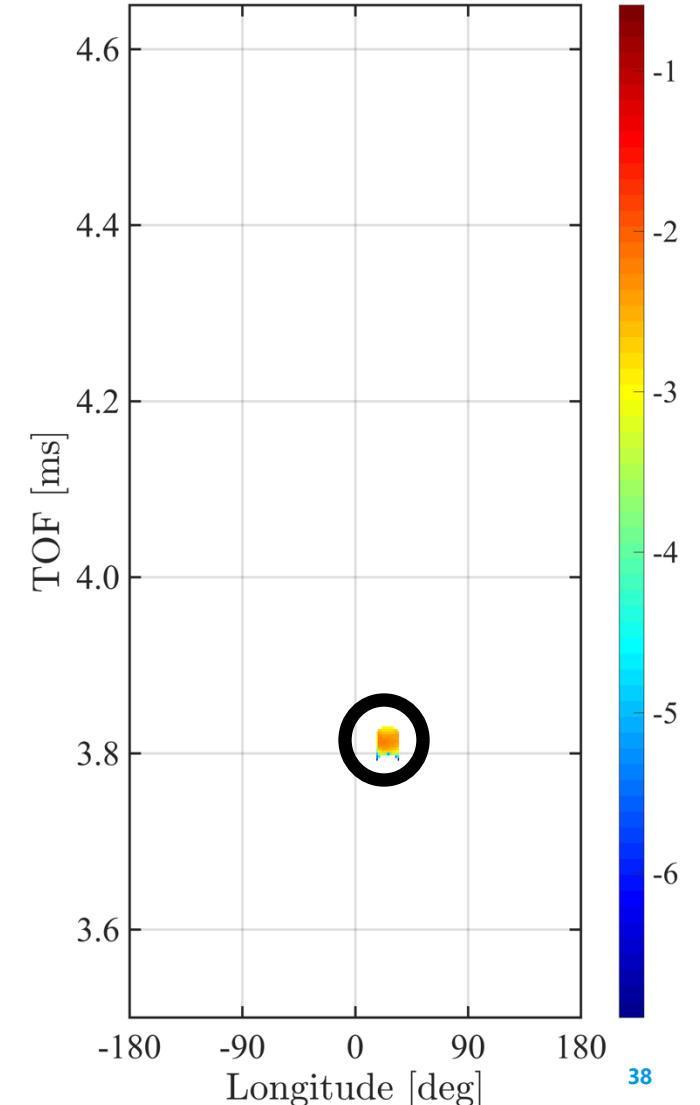
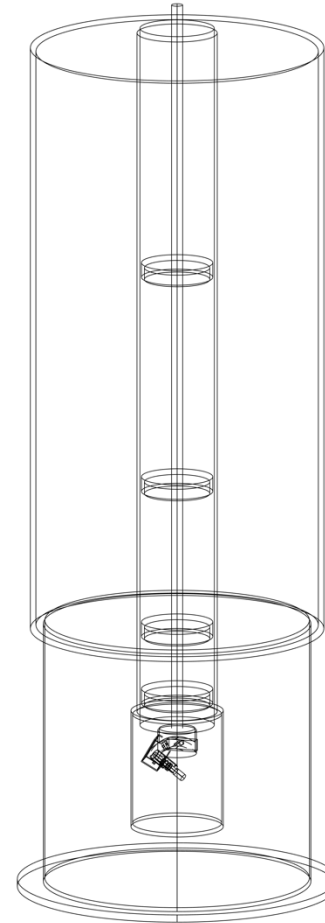
- Wish to understand how individual background features arose



Conditional components

Limiting logger components to record interesting neutrons

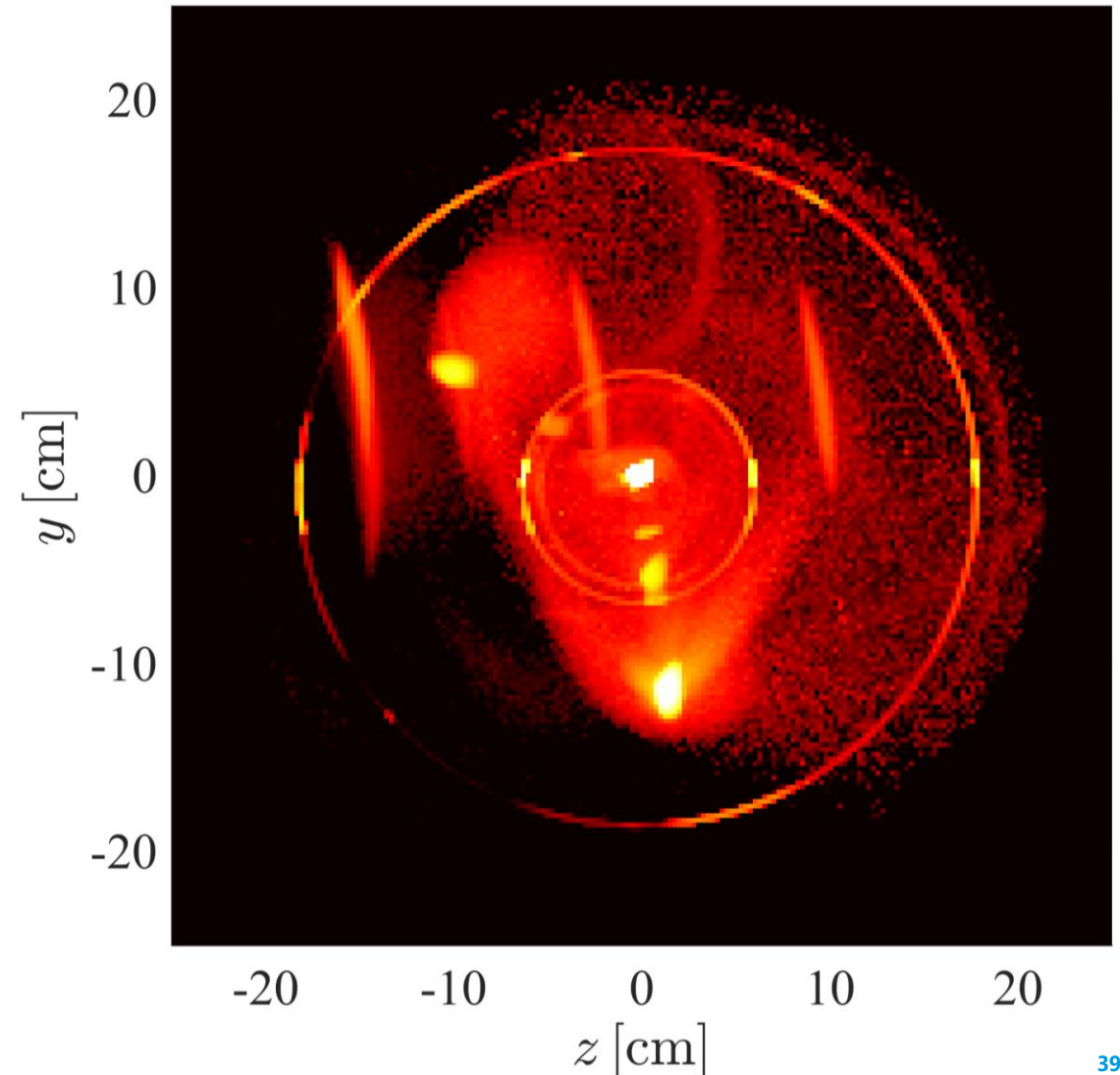
- Wish to understand how individual background features arose
- Select subset of output to be recorded in loggers using Union conditionals



Conditional components

Limiting logger components to record interesting neutrons

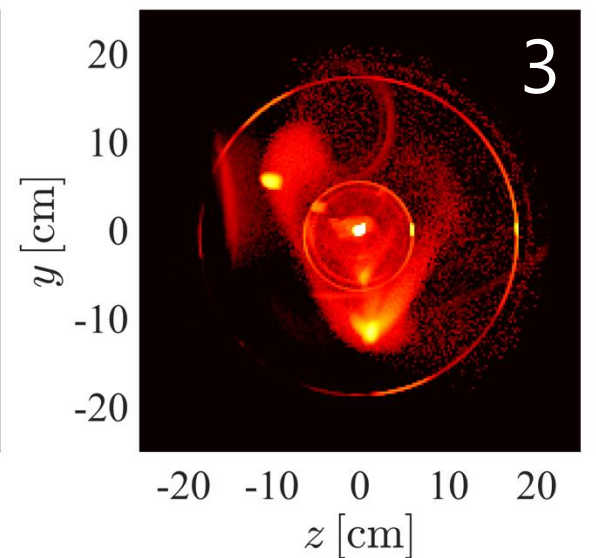
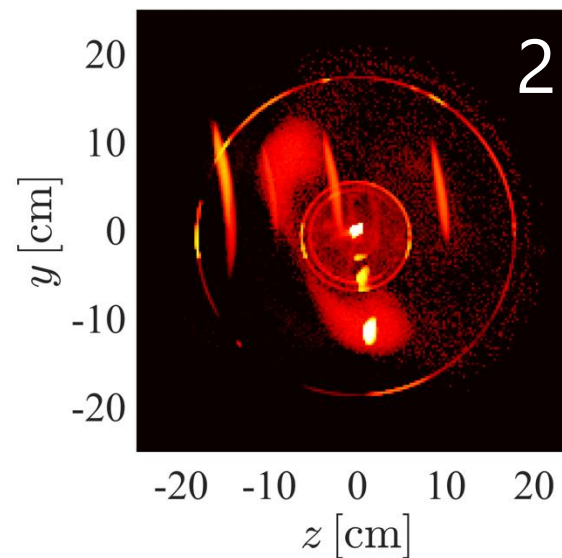
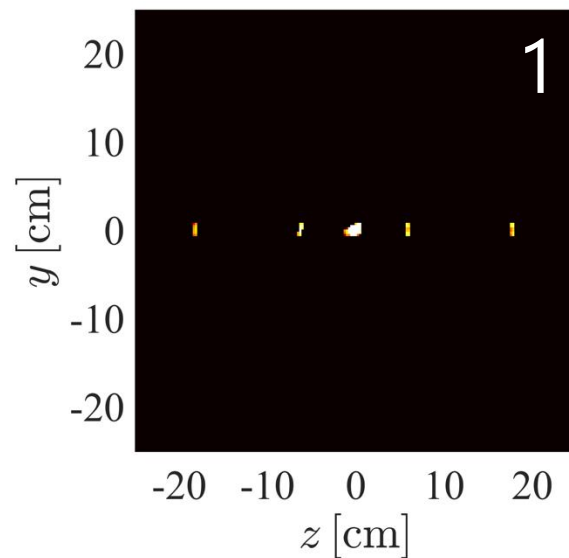
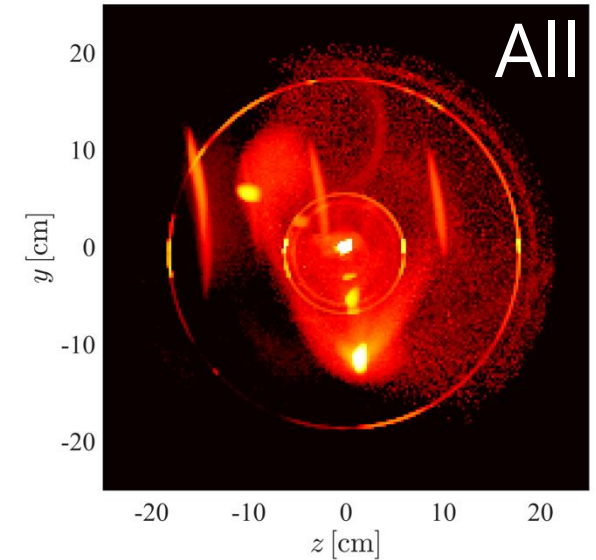
- Can limit the events that are recorded to the logger using conditionals
- Here only rays that contribute to the chosen background problem is seen
- Spatial distribution of scattered intensity



Conditional components

Limiting logger components to record interesting neutrons

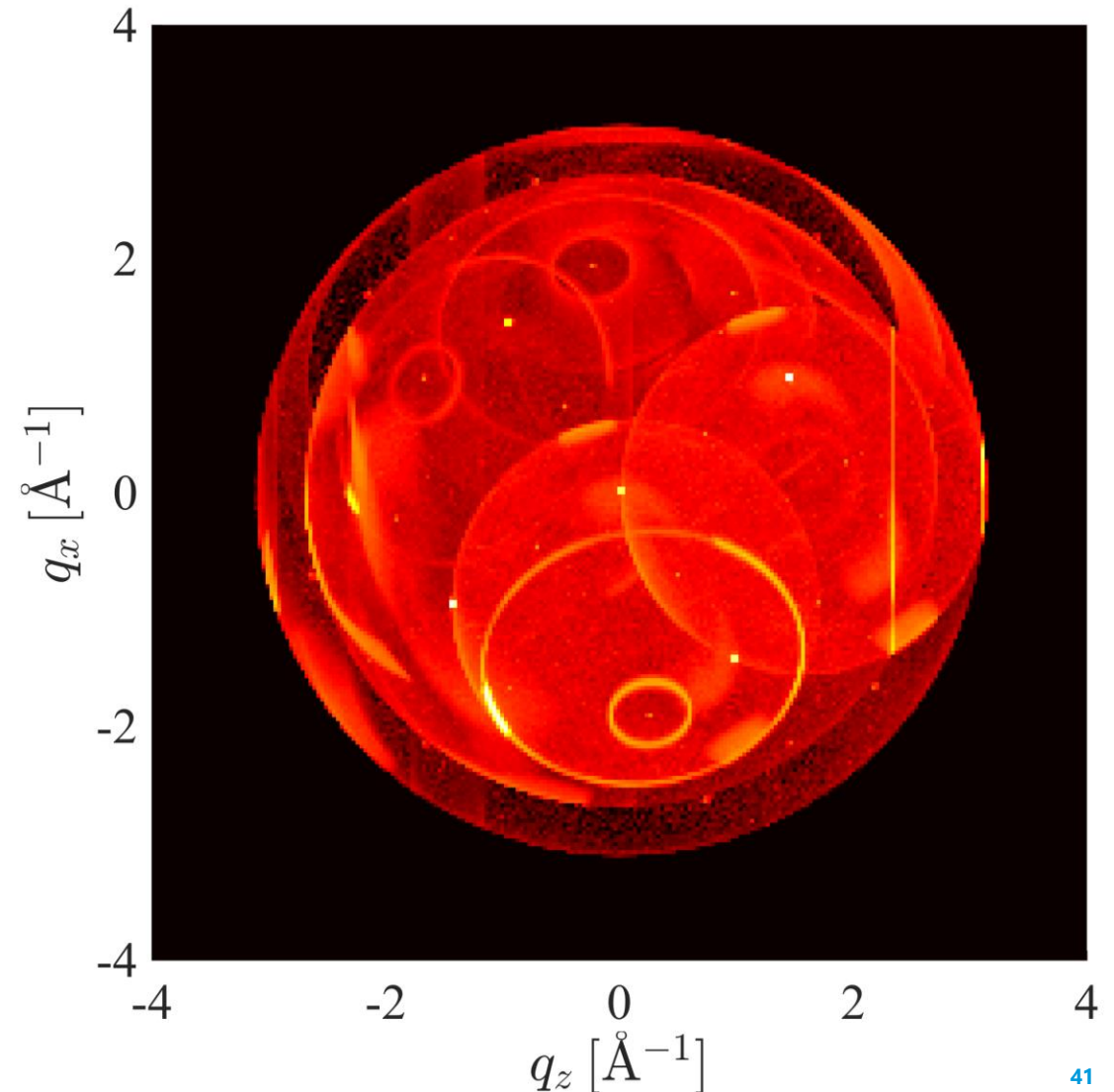
- The different scattering orders makes it easier to draw conclusions
- The sample is clearly involved



Conditional components

Limiting logger components to record interesting neutrons

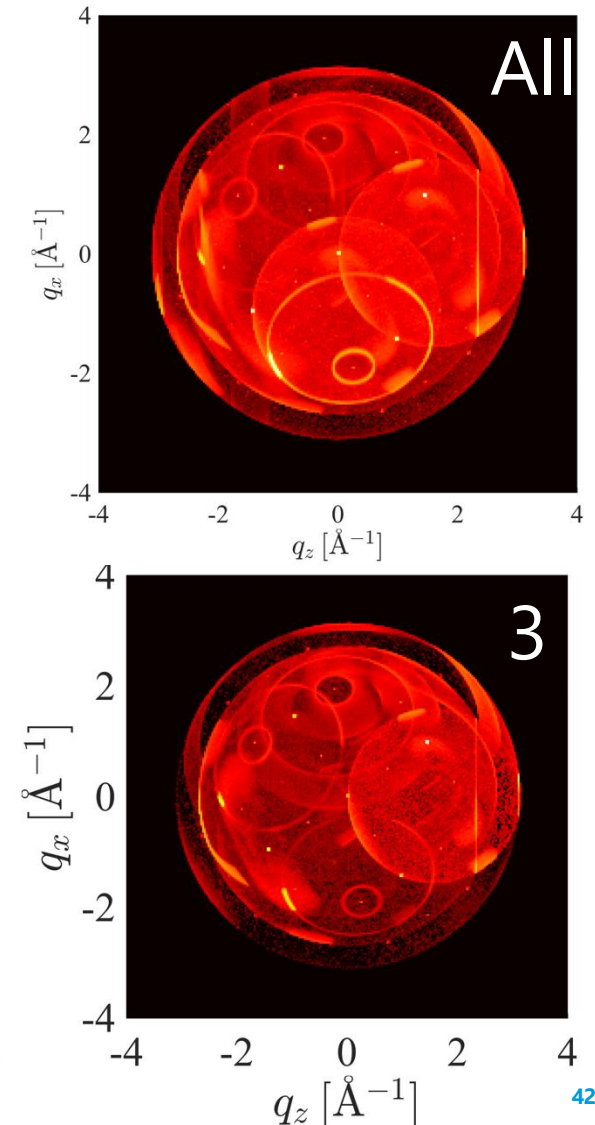
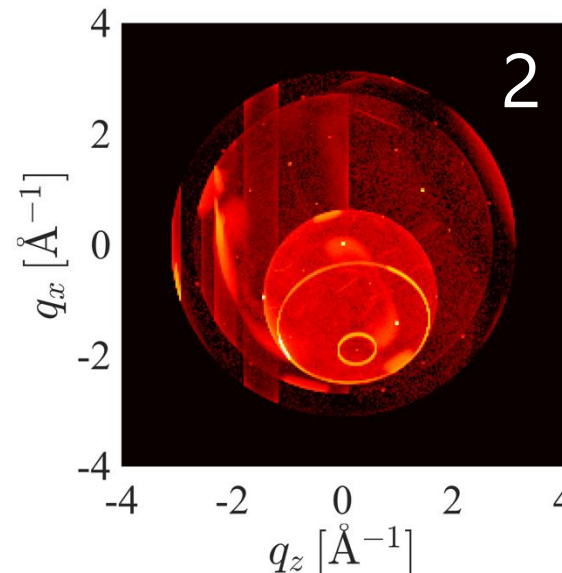
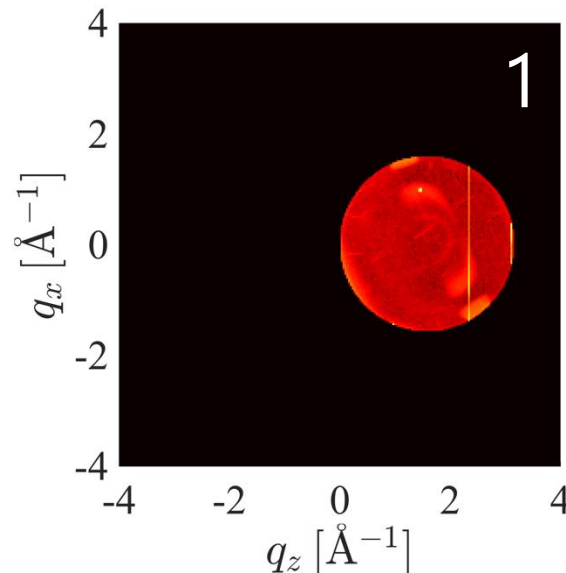
- Scattering vectors of rays that contribute to the chosen background problem

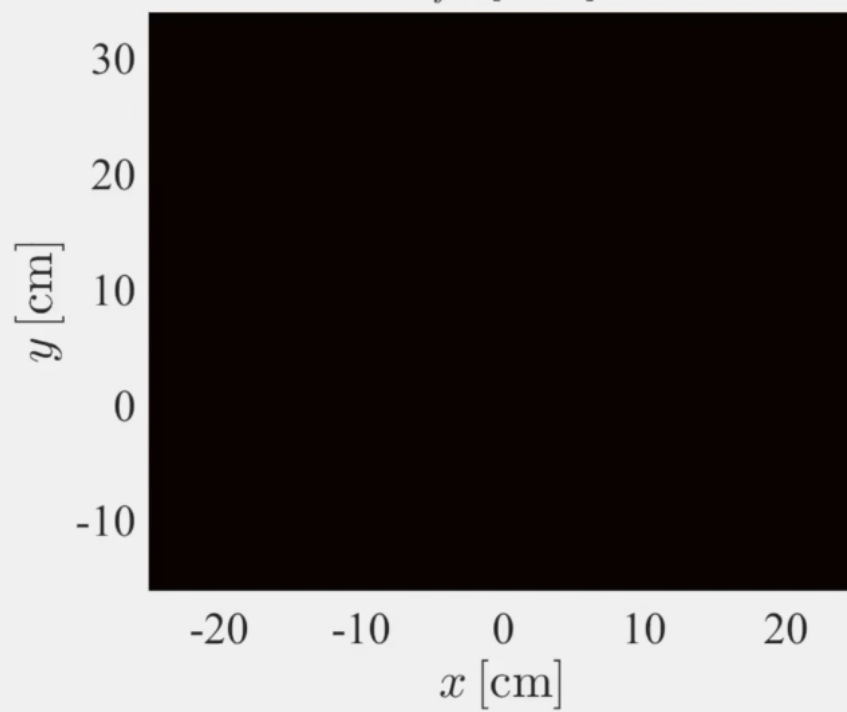
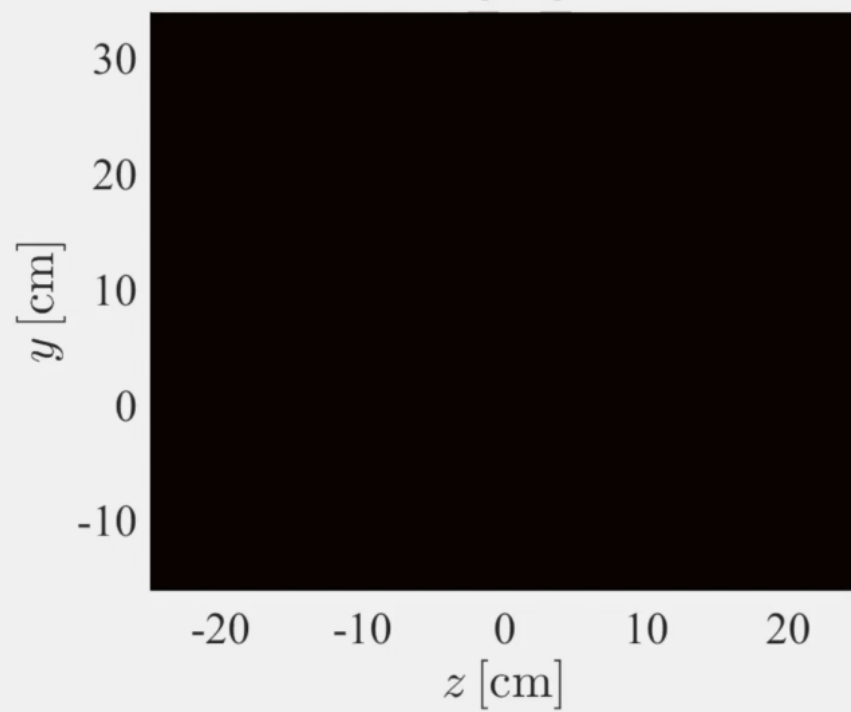
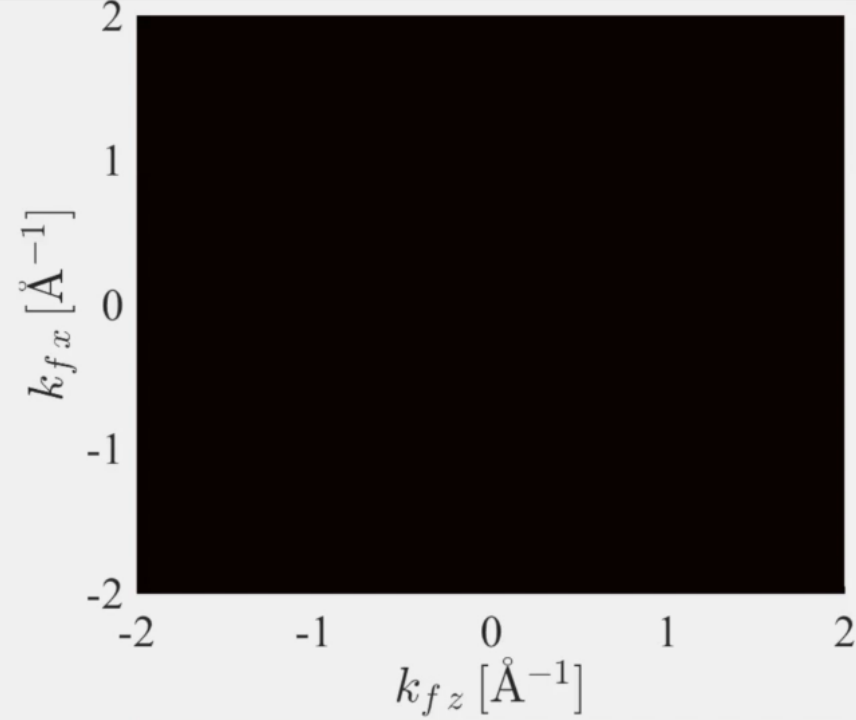
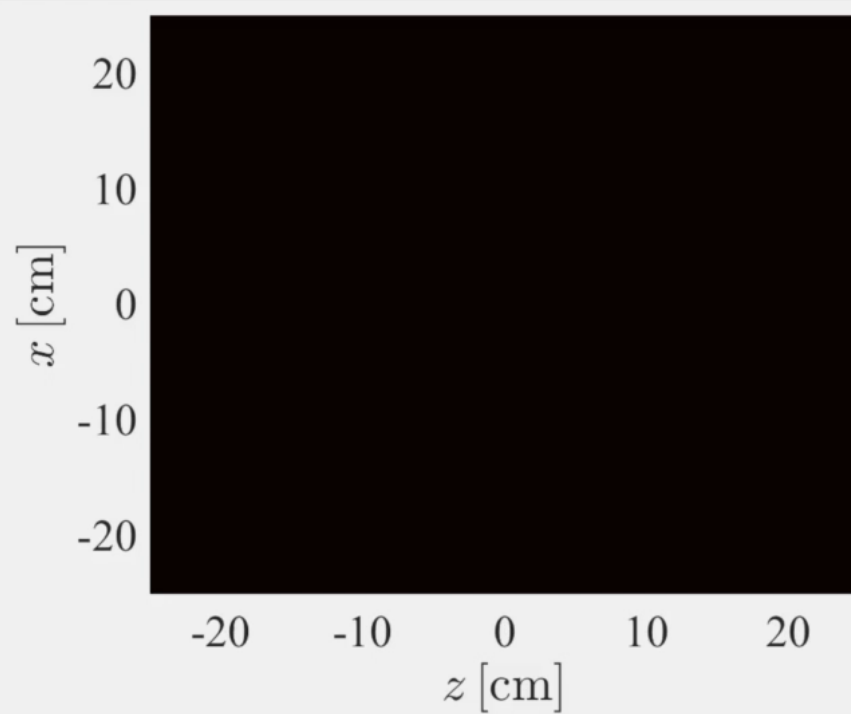


Conditional components

Limiting logger components to record interesting neutrons

- The different scattering orders makes it easier to draw conclusions
- Single crystal Bragg peak to powder Debye Scherrer cone







Summary

- Union components allow simulation of complex physics and geometry
- Full multiple scattering simulation provides surprising level of detail
- The modular nature makes it easier to expand
 - New geometry with a geometry component (Master project contributed cone and CAD)
 - New process component to add new physics
- Full tutorial available in McStasScript format!

Thanks for your attention



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